KU LEUVEN

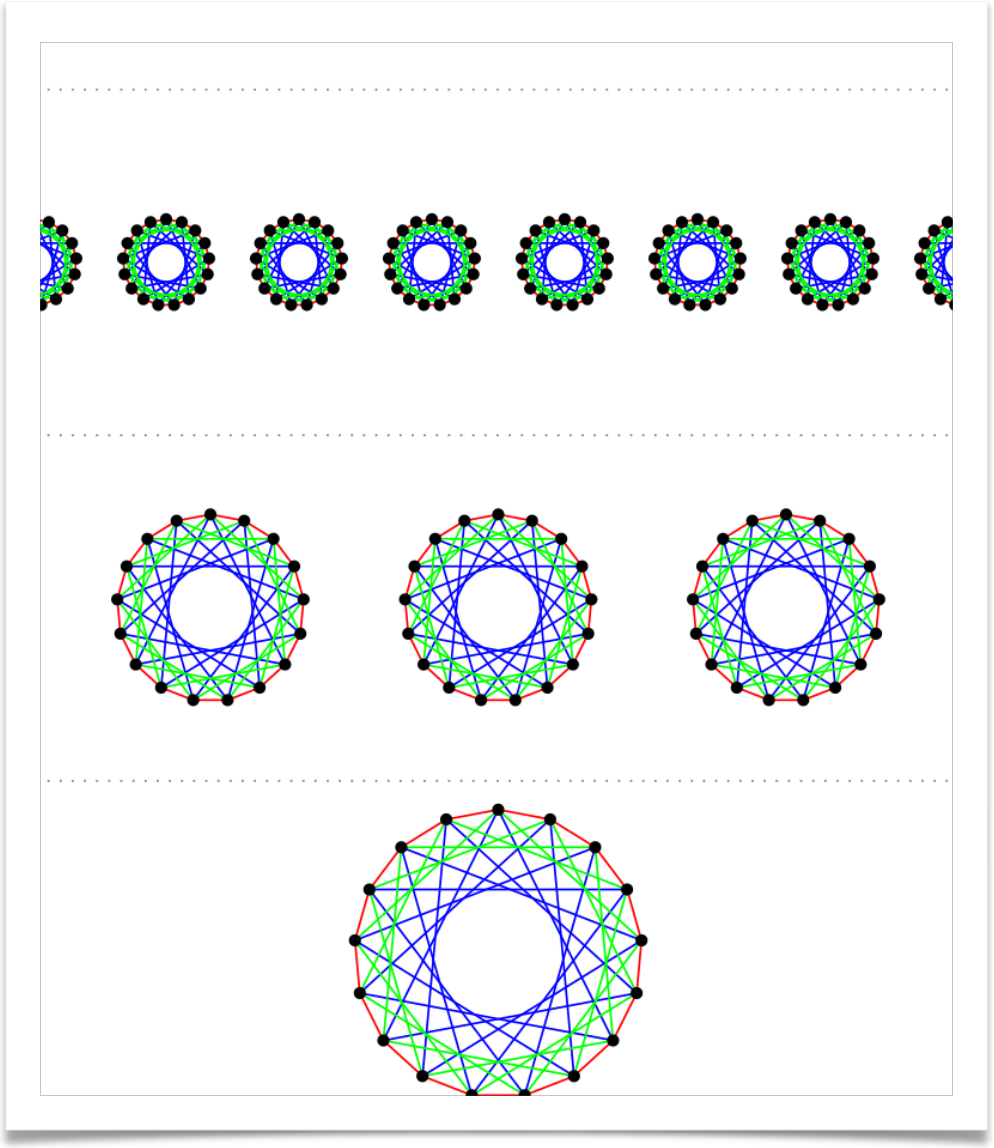
MIKE: An efficient and compact NIKE Based on a Commutative Monoidal Action

First part: joint work with Emil Olaisen

The rest: joint work with the MIKE team
(Andrea Basso, Pierrick Dartois, Max Duparc, Jonathan Komada Eriksen, Sabrina Kunzweiler, Michael Meyer, Giacomo Pope, Krijn Reijnders, Damien Robert, Ryan Rueger and Sina Alexandra Schaeffler)

Graded monoidal actions

Not as scary as it sounds



Disclaimer

This talk:

Graded monoidal actions → **Module-action/ MIKE**

Reality:

Module-action/ MIKE → **Graded monoidal actions**

Rationale: Graded monoidal actions are easy to understand and give a good motivation for MIKE!

Historical development

Finite field DH

KU LEUVEN

The Diffie-Hellman KEX v.1*

Public parameters:

p**, big prime,**

$g \in \mathrm{Gen}(\mathbb{F}_p^\times)$**, a generator of** $(\mathbb{F}_p)^\times$

Alice: **Bob:**

*like most other cults, we will shamelessly rewrite history to fit our narrative

The Diffie-Hellman KEX v.1*

Public parameters:

p**, big prime,**

$g \in \mathrm{Gen}(\mathbb{F}_p^\times)$**, a generator of** $(\mathbb{F}_p)^\times$

Alice:

sk: $s_A \in (\mathbb{Z}/(p-1)\mathbb{Z})^\times$

pk: $A := g^{s_A} \in \mathrm{Gen}(\mathbb{F}_p^\times)$

Bob:

*like most other cults, we will shamelessly rewrite history to fit our narrative

The Diffie-Hellman KEX v.1*

Public parameters:

p, **big prime,**

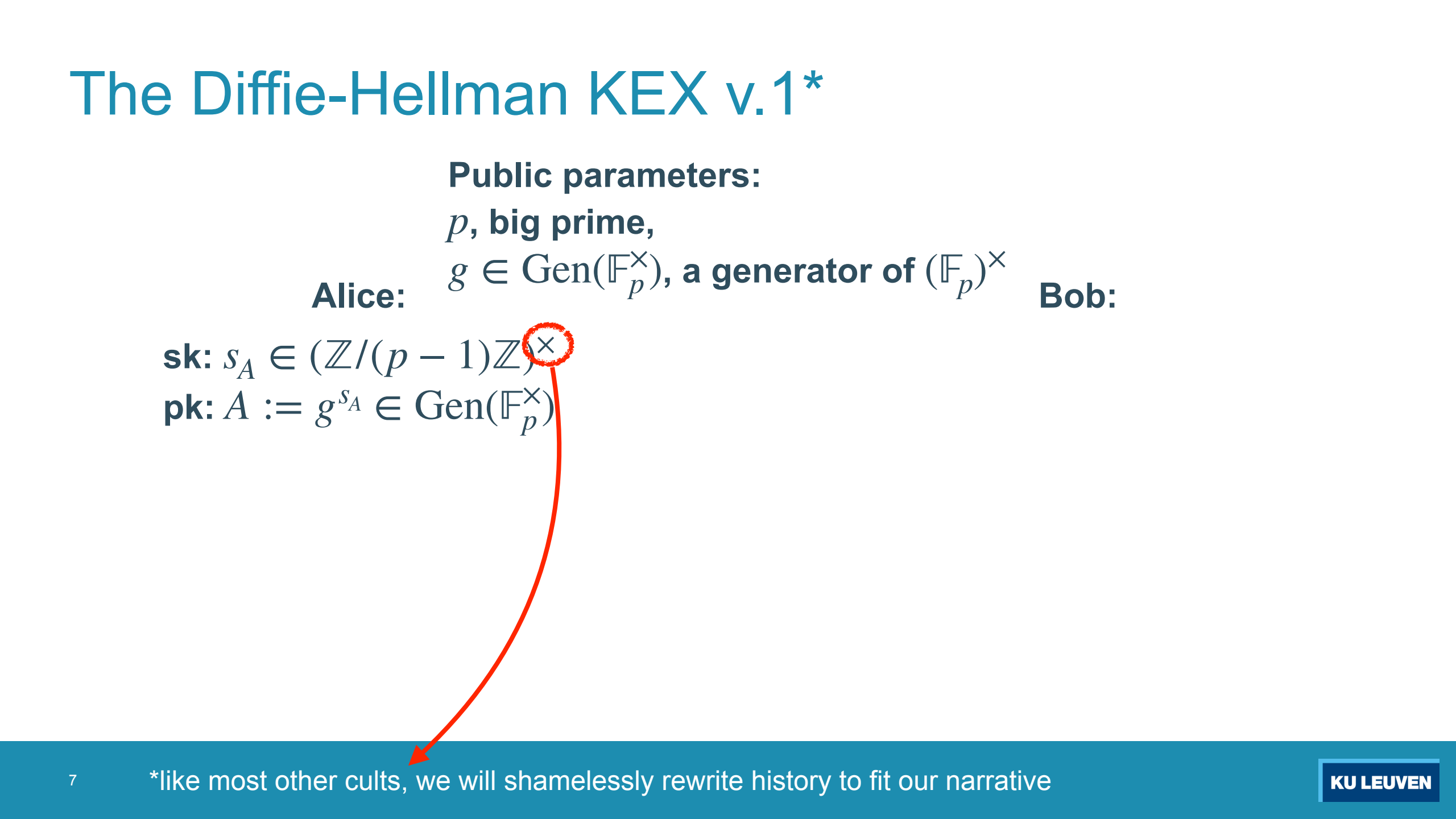
$g \in \mathrm{Gen}(\mathbb{F}_p^\times)$, **a generator of** $(\mathbb{F}_p)^\times$

Alice:

sk: $s_A \in (\mathbb{Z}/(p-1)\mathbb{Z})^\times$

pk: $A := g^{s_A} \in \mathrm{Gen}(\mathbb{F}_p^\times)$

Bob:



*like most other cults, we will shamelessly rewrite history to fit our narrative

The Diffie-Hellman KEX v.1*

Public parameters:

p**, big prime,**

$g \in \mathrm{Gen}(\mathbb{F}_p^\times)$**, a generator of** $(\mathbb{F}_p)^\times$

Alice:

sk: $s_A \in (\mathbb{Z}/(p-1)\mathbb{Z})^\times$

pk: $A := g^{s_A} \in \mathrm{Gen}(\mathbb{F}_p^\times)$

Bob:

sk: $s_B \in (\mathbb{Z}/(p-1)\mathbb{Z})^\times$

pk: $B := g^{s_B} \in \mathrm{Gen}(\mathbb{F}_p^\times)$

*like most other cults, we will shamelessly rewrite history to fit our narrative

The Diffie-Hellman KEX v.1*

Public parameters:

p, **big prime,**

$g \in \mathrm{Gen}(\mathbb{F}_p^\times)$, **a generator of** $(\mathbb{F}_p)^\times$

Alice:

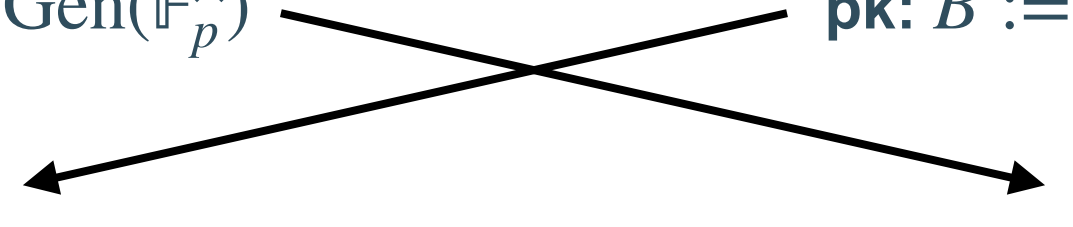
sk: $s_A \in (\mathbb{Z}/(p-1)\mathbb{Z})^\times$

pk: $A := g^{s_A} \in \mathrm{Gen}(\mathbb{F}_p^\times)$

Bob:

sk: $s_B \in (\mathbb{Z}/(p-1)\mathbb{Z})^\times$

pk: $B := g^{s_B} \in \mathrm{Gen}(\mathbb{F}_p^\times)$



*like most other cults, we will shamelessly rewrite history to fit our narrative

The Diffie-Hellman KEX v.1*

Public parameters:

p, **big prime,**

$g \in \mathrm{Gen}(\mathbb{F}_p^\times)$, **a generator of** $(\mathbb{F}_p)^\times$

Alice:

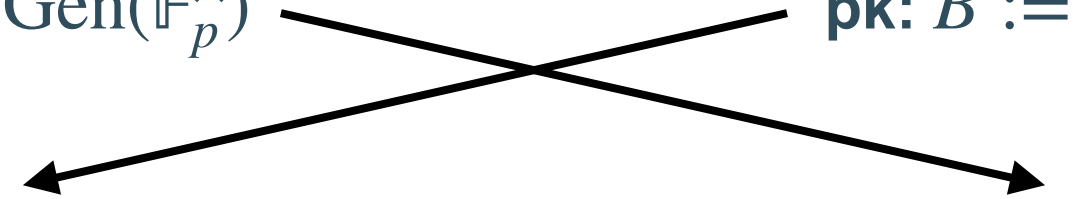
sk: $s_A \in (\mathbb{Z}/(p-1)\mathbb{Z})^\times$

pk: $A := g^{s_A} \in \mathrm{Gen}(\mathbb{F}_p^\times)$

Bob:

sk: $s_B \in (\mathbb{Z}/(p-1)\mathbb{Z})^\times$

pk: $B := g^{s_B} \in \mathrm{Gen}(\mathbb{F}_p^\times)$



B A

ss: $B^{s_A} \in \mathrm{Gen}(\mathbb{F}_p^\times)$

ss: $A^{s_B} \in \mathrm{Gen}(\mathbb{F}_p^\times)$

*like most other cults, we will shamelessly rewrite history to fit our narrative

The Diffie-Hellman KEX v.1*

Public parameters:

p, **big prime,**

$g \in \mathrm{Gen}(\mathbb{F}_p^\times)$, **a generator of** $(\mathbb{F}_p)^\times$

Alice:

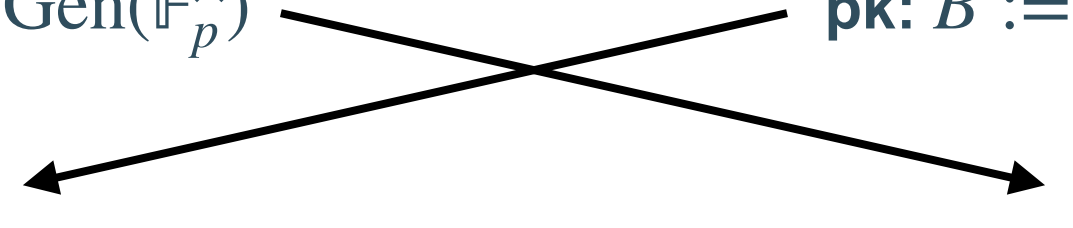
sk: $s_A \in (\mathbb{Z}/(p-1)\mathbb{Z})^\times$

pk: $A := g^{s_A} \in \mathrm{Gen}(\mathbb{F}_p^\times)$

Bob:

sk: $s_B \in (\mathbb{Z}/(p-1)\mathbb{Z})^\times$

pk: $B := g^{s_B} \in \mathrm{Gen}(\mathbb{F}_p^\times)$



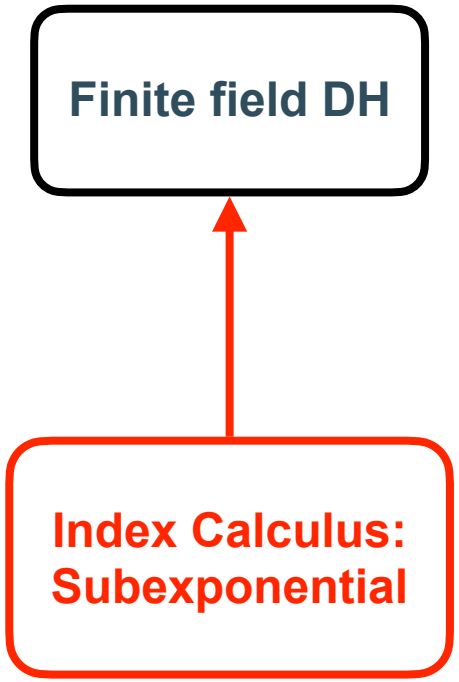
ss: $B^{s_A} \in \mathrm{Gen}(\mathbb{F}_p^\times)$

ss: $A^{s_B} \in \mathrm{Gen}(\mathbb{F}_p^\times)$

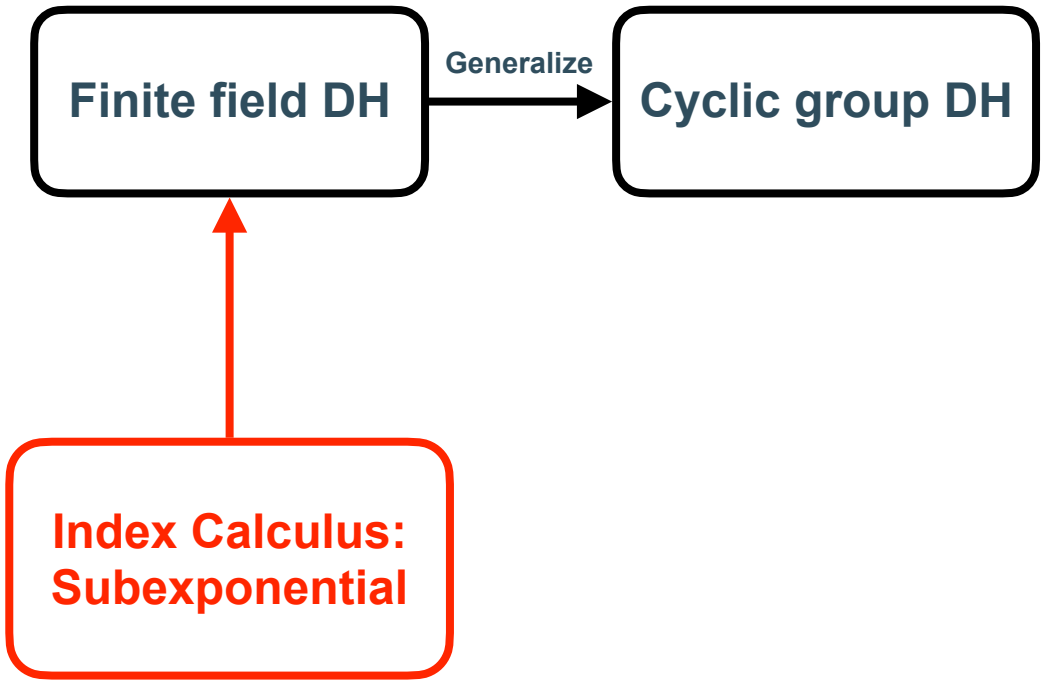
$$B^{s_A} = g^{s_B s_A} = g^{s_A s_B} = A^{s_B}$$

*like most other cults, we will shamelessly rewrite history to fit our narrative

Historical development



Historical development



The Diffie-Hellman KEX v.2*

Public parameters:

H, **cyclic group of order** n

$x_0 \in \mathrm{Gen}(H)$, **a generator of** H

Alice:

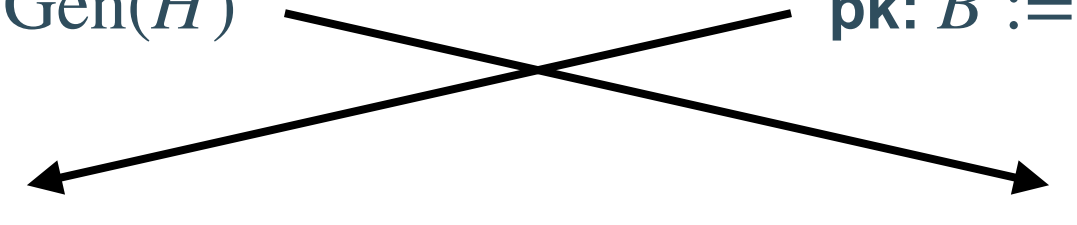
sk: $s_A \in (\mathbb{Z}/n\mathbb{Z})^\times$

pk: $A := x_0^{s_A} \in \mathrm{Gen}(H)$

Bob:

sk: $s_B \in (\mathbb{Z}/n\mathbb{Z})^\times$

pk: $B := x_0^{s_B} \in \mathrm{Gen}(H)$



B A

ss: $B^{s_A} \in \mathrm{Gen}(H)$

ss: $A^{s_B} \in \mathrm{Gen}(H)$

$$B^{s_A} = x_0^{s_B s_A} = x_0^{s_A s_B} = A^{s_B}$$

*like most other cults, we will shamelessly rewrite history to fit our narrative

The Diffie-Hellman KEX v.2*

Public parameters:

H, **cyclic group of order** n

$x_0 \in \mathrm{Gen}(H)$, **a generator of** H

Alice:

sk: $s_A \in (\mathbb{Z}/n\mathbb{Z})^\times$

pk: $A := x_0^{s_A} \in \mathrm{Gen}(H)$

sk: s_B

pk: B

B

ss: $B^{s_A} \in \mathrm{Gen}(H)$

ss: $A^{s_B} \in \mathrm{Gen}(H)$

V2. Specialises to v1. by

$H = (\mathbb{F}_p)^\times$

Other popular choices:

$H = \langle g \rangle$, **for some**

$g \in (\mathbb{F}_p)^\times$

(not necessarily a generator)

Or of course:

$H = \langle P \rangle, P \in E$

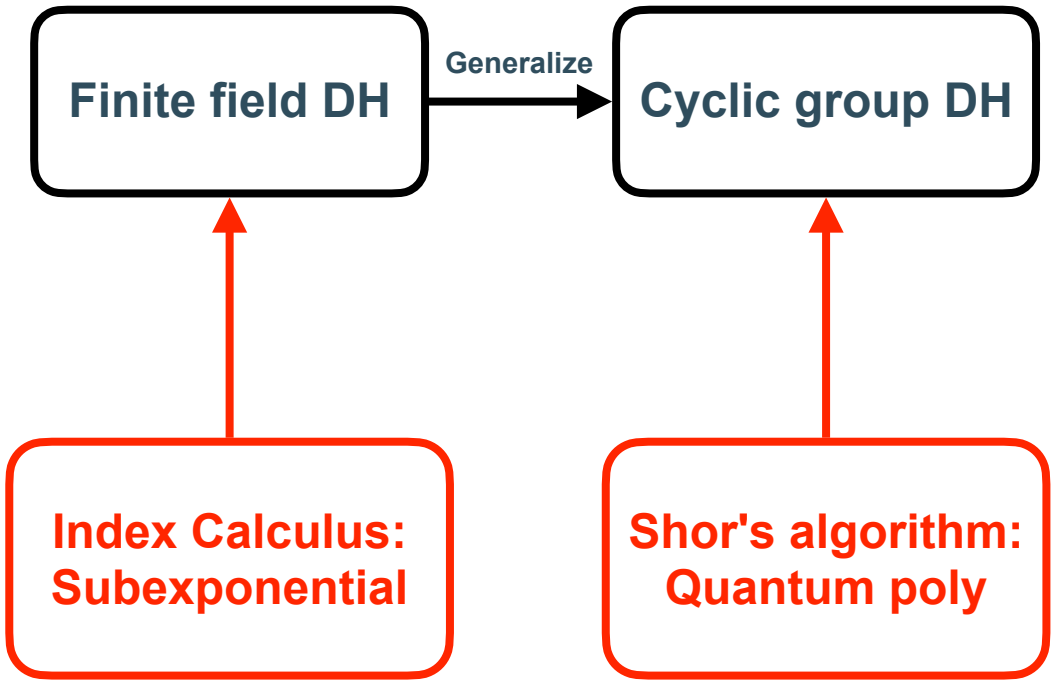
for some elliptic curve E

$$B^{s_A} = x_0^{s_B s_A} = x_0^{s_A s_B} = A^{s_B}$$

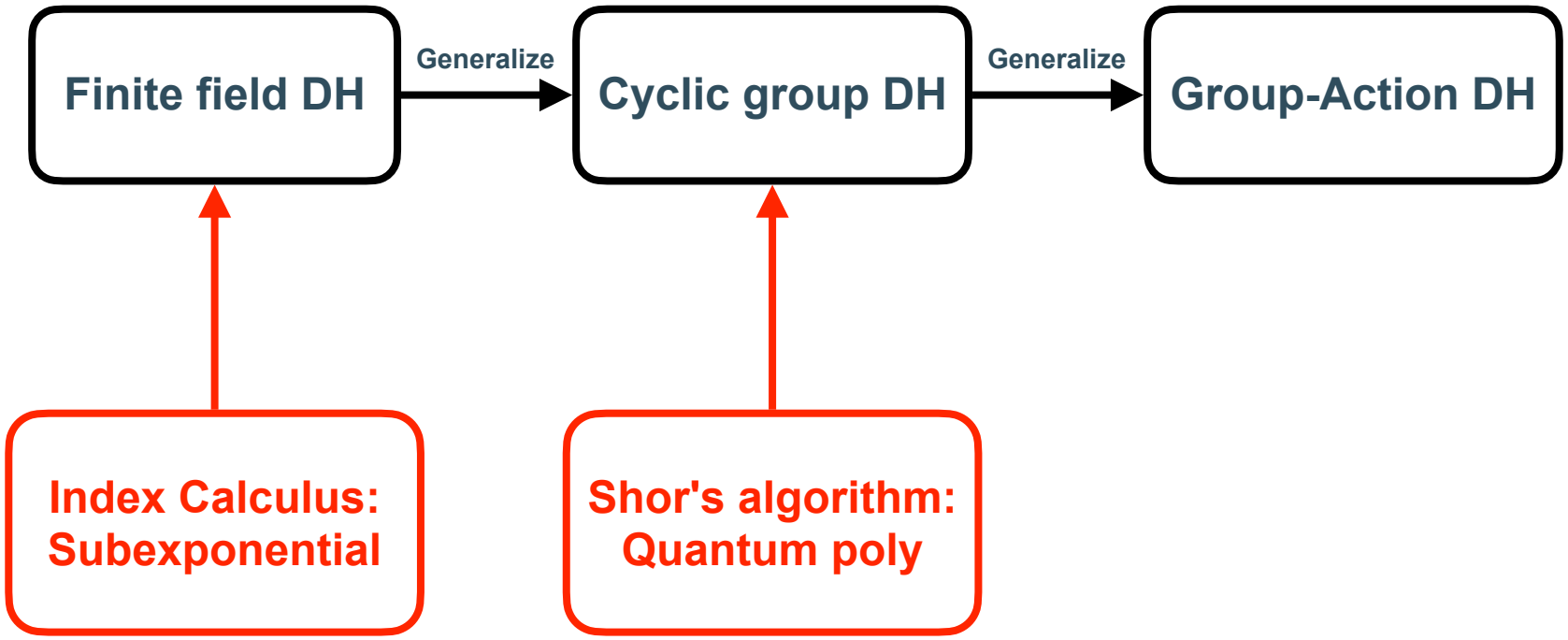
*like most other cults, we will shamelessly rewrite history to fit our narrative

KU LEUVEN

Historical development



Historical development



The Diffie-Hellman KEX v.3

Public parameters:

$\star : G \times X \to X$, **regular ab. group action of order** n

fixed element $x_0 \in X$

Alice:

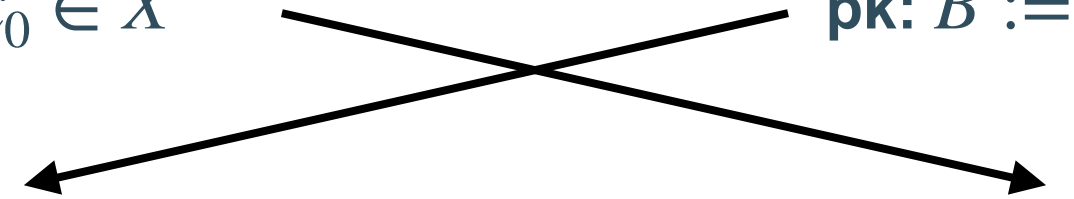
sk: $s_A \in G$

pk: $A := s_A \star x_0 \in X$

Bob:

sk: $s_B \in G$

pk: $B := s_B \star x_0 \in X$



B A

ss: $s_A \star B \in X$

ss: $s_B \star A \in X$

$$s_A \star B = (s_B s_A) \star x_0 = (s_A s_B) \star x_0 = s_B \star A$$

The Diffie-Hellman KEX v.3

Public parameters:

$\star : G \times X \to X$, **regular ab. g**

fixed element $x_0 \in X$

V3. Specialises to v2. by

$G = (\mathbb{Z}/n\mathbb{Z})^\times,$

$X = \mathrm{Gen}(H)$

Other popular choices:

$G = Cl(R),$

$X = Ell_p(R)$

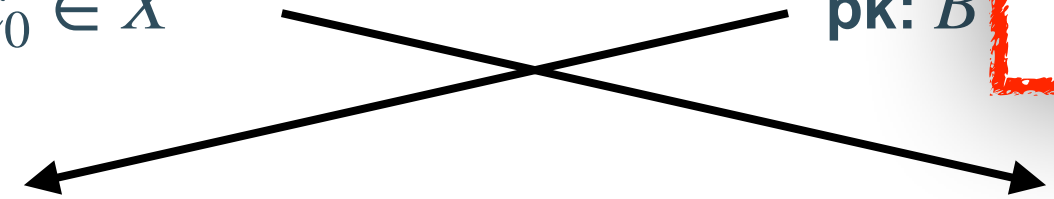
Alice:

sk: $s_A \in G$

pk: $A := s_A \star x_0 \in X$

sk: s_B

pk: B



B

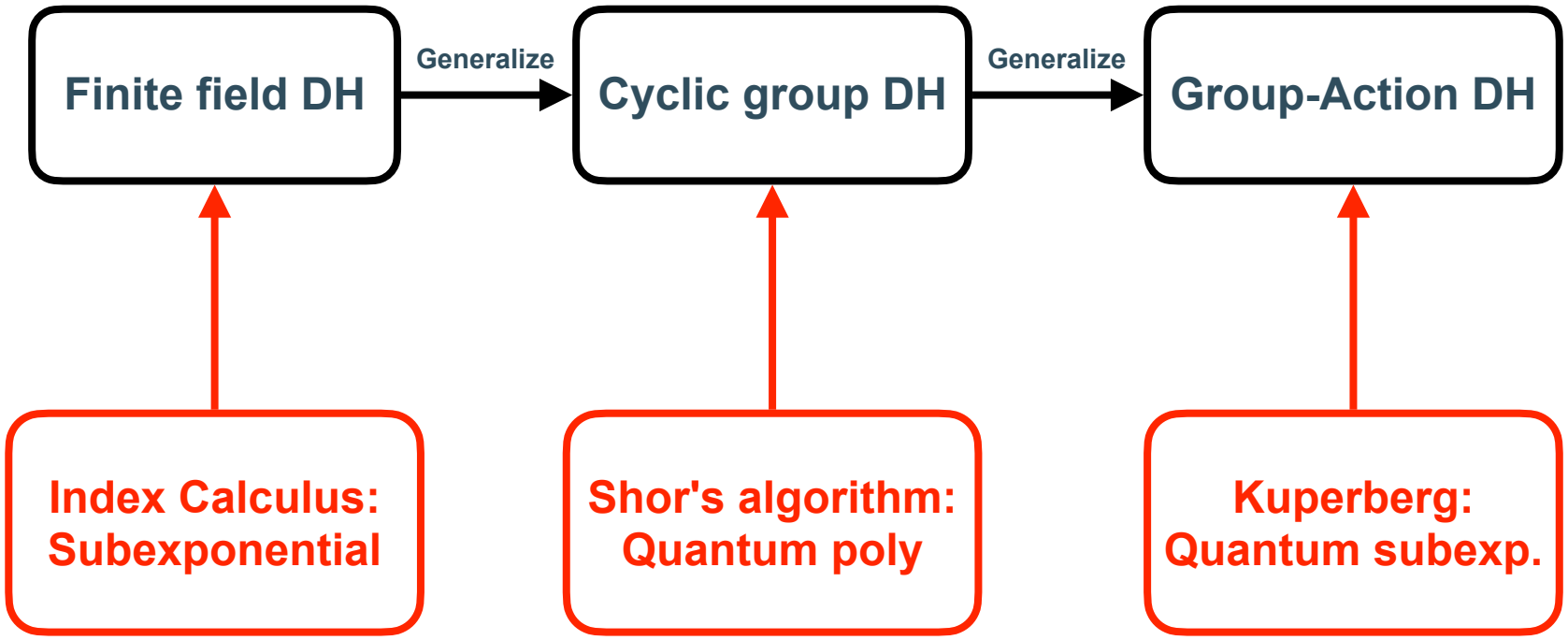
A

ss: $s_A \star B \in X$

ss: $s_B \star A \in X$

$$s_A \star B = (s_B s_A) \star x_0 = (s_A s_B) \star x_0 = s_B \star A$$

Historical development



Historical development



Finite field DH —Generalize→ Cyclic group DH —Generalize→ Group-Action DH —Generalize→

Index Calculus: Subexponential

Shor's algorithm: Quantum poly

Kuperberg: Quantum subexp.

Graded monoid(al actions)

NB! every monoid will be assumed to be commutative

KU LEUVEN

Graded monoid(al actions)

Recall (informally): "A monoid $(M, \otimes)$ is like a group without inverses"

Examples:

- $(\mathbb{N}, \cdot)$, $(\mathbb{Z}_{\geq 0}, +)$
- any ring R is (by definition) a monoid under multiplication

Graded monoid(al actions)

Recall (informally): "A monoid $(M, \otimes)$ is like a group without inverses"

A (multiplicatively) graded monoid M is a monoid for which:

- we can write $M = \sqcup_{i \in \mathbb{N}} M_i$

- $M_r \otimes M_s \subseteq M_{rs}$

Examples:

- $(\mathbb{N}, \cdot)$, $(\mathbb{Z}_{\geq 0}, +)$
- any ring R is (by definition) a monoid under multiplication

Example (additive):

- $(\mathbb{Q}[X] \setminus \{0\}, \cdot)$, grading given by degree

KU LEUVEN

Graded monoid(al actions)

Recall (informally): "A monoid $(M, \otimes)$ is like a group without inverses"

A (multiplicatively) graded monoid M is a monoid for which:

- **we can write** $M = \sqcup_{i \in \mathbb{N}} M_i$
- $M_r \otimes M_s \subseteq M_{rs}$

Examples:

- $(\mathbb{N}, \cdot)$, $(\mathbb{Z}_{\geq 0}, +)$
- any ring R is (by definition) a monoid under multiplication

Example (additive):

- $(\mathbb{Q}[X] \setminus \{0\}, \cdot)$, grading given by degree

Succinctly: A monoid together with a homomorphism $\mathrm{rnk} : M \to (\mathbb{N}, \cdot)$

KU LEUVEN

Graded monoidal actions

A (multiplicatively) graded monoidal action $(M, X, \star)$ is

- **A monoidal action**
- **M is a multiplicatively graded monoid**
- **We can write $X = \sqcup_{i \in \mathbb{N}} X_i$**
- $M_r \star X_s \subseteq X_{rs}$

Graded monoidal actions

A (multiplicatively) graded monoidal action $(M, X, \star)$ is

- **A monoidal action**
- **M is a multiplicatively graded monoid**
- **We can write $X = \sqcup_{i \in \mathbb{N}} X_i$**
- $M_r \star X_s \subseteq X_{rs}$

Important example:

- Let $(G, X, \star)$ be any group action.

Then

$$\star' : (\mathbb{N} \times G) \times (\mathbb{N} \times X) \to (\mathbb{N} \times X)$$
$$((n, g), (m, x)) \to (nm, g \star x)$$

is a graded monoidal action, since $\mathbb{N} \times G = \sqcup_{i \in \mathbb{N}} (i, G)$ and $\mathbb{N} \times X = \sqcup_{i \in \mathbb{N}} (i, X)$

KU LEUVEN

Finite, free and transitive?

Recall: For cryptographic applications, group actions $(G, X, \star)$ should be finite, transitive and free.

Finite, free and transitive?

Recall: For cryptographic applications, group actions $(G, X, \star)$ should be finite, transitive and free.



There is a (actually, many) bijection(s) $G \leftrightarrow X$ of finite sets.

Finite, free and transitive?

Recall: For cryptographic applications, group actions $(G, X, \star)$ should be finite, transitive and free.



There is a (actually, many) bijection(s) $G \leftrightarrow X$ of finite sets.

Generalises?:

- **Finite:**
- **Transitive:**
- **Free:**

Finite, free and transitive?

Recall: For cryptographic applications, group actions $(G, X, \star)$ should be finite, transitive and free.



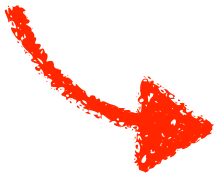
There is a (actually, many) bijection(s) $G \leftrightarrow X$ of finite sets.

Generalises?:

- **Finite: ✗ for $m \in M_{\geq 2}$, $\{m, m \otimes m, m \otimes m \otimes m, \ldots\}$ is infinite.**
- **Transitive:**
- **Free:**

Finite, free and transitive?

Recall: For cryptographic applications, group actions $(G, X, \star)$ should be finite, transitive and free.



There is a (actually, many) bijection(s) $G \leftrightarrow X$ of finite sets.

Generalises?:

- **Finite: ✗ for $m \in M_{\geq 2}$, $\{m, m \otimes m, m \otimes m \otimes m, \ldots\}$ is infinite.**
- **Transitive: ✗ we can never go down in dimension.**
- **Free:**

Finite, free and transitive?

Recall: For cryptographic applications, group actions $(G, X, \star)$ should be finite, transitive and free.



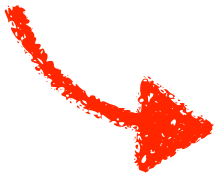
There is a (actually, many) bijection(s) $G \leftrightarrow X$ of finite sets.

Generalises?:

- **Finite: ✗ for $m \in M_{\geq 2}$, $\{m, m \otimes m, m \otimes m \otimes m, \ldots\}$ is infinite.**
- **Transitive: ✗ we can never go down in dimension.**
- **Free: ? Possible, but too strong (implies M is cancellable).**

Finite, free and transitive?

Recall: For cryptographic applications, group actions $(G, X, \star)$ should be finite, transitive and free.



There is a (actually, many) bijection(s) $G \leftrightarrow X$ of finite sets.

Generalises?:

- **Finite: ✗ for $m \in M_{\geq 2}$, $\{m, m \otimes m, m \otimes m \otimes m, \ldots\}$ is infinite.**
- **Transitive: ✗ we can never go down in dimension.**
- **Free: ? Possible, but too strong (implies M is cancellable).**

New goal: Axioms to recover finite bijection between M_r and X_r

Finite, free and transitive?

Let M be a graded monoid, and $(M, X, \star)$ a graded monoidal action

- **Rankwise finite:**
- **Base-transitive:**
- **Base-free:**

KU LEUVEN

Finite, free and transitive?

Let M be a graded monoid, and $(M, X, \star)$ a graded monoidal action

- **Rankwise finite: M_d are all finite**
- **Base-transitive:**
- **Base-free:**

KU LEUVEN

Finite, free and transitive?

Let M be a graded monoid, and $(M, X, \star)$ a graded monoidal action

- **Rankwise finite: M_d are all finite**
- **Base-transitive: For all $x \in X_1, y \in X$ there exists $a \in M$ s.t. $a \star x = y$**
- **Base-free:**

KU LEUVEN

Finite, free and transitive?

Let M be a graded monoid, and $(M, X, \star)$ a graded monoidal action

- **Rankwise finite: M_d are all finite**
- **Base-transitive: For all $x \in X_1, y \in X$ there exists $a \in M$ s.t. $a \star x = y$**
- **Base-free: For all $x \in X_1$ and $a, b \in M$, $a \star x = b \star x$ implies that $a = b$**

KU LEUVEN

Finite, free and transitive?

Let M be a graded monoid, and $(M, X, \star)$ a graded monoidal action

- **Rankwise finite: M_d are all finite**
- **Base-transitive: For all $x \in X_1, y \in X$ there exists $a \in M$ s.t. $a \star x = y$**
- **Base-free: For all $x \in X_1$ and $a, b \in M$, $a \star x = b \star x$ implies that $a = b$**

Consequence 1: $M_r \leftrightarrow X_r$ bijection of finite sets

KU LEUVEN

Finite, free and transitive?

Let M be a graded monoid, and $(M, X, \star)$ a graded monoidal action

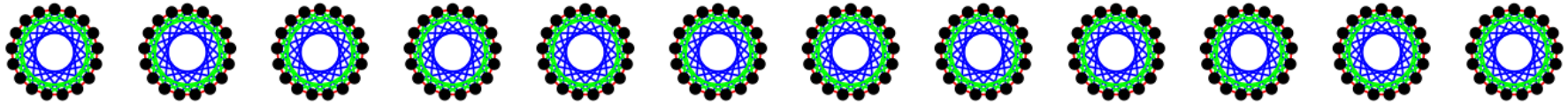
- **Rankwise finite: M_d are all finite**
- **Base-transitive: For all $x \in X_1, y \in X$ there exists $a \in M$ s.t. $a \star x = y$**
- **Base-free: For all $x \in X_1$ and $a, b \in M$, $a \star x = b \star x$ implies that $a = b$**

Consequence 1: $M_r \leftrightarrow X_r$ bijection of finite sets

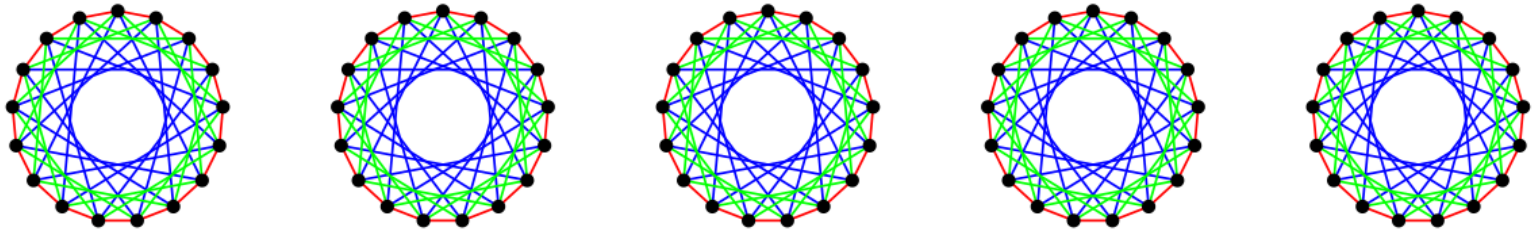
Consequence 2: M_1 is a group, and $(M_1, X_1, \star_{1,1})$ is a regular group action

dimension ≥ 4

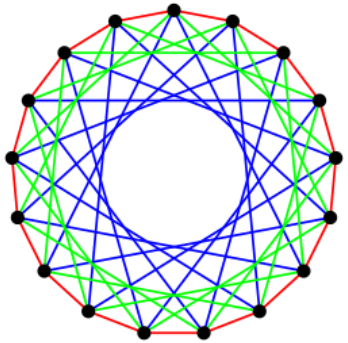
dimension 3



dimension 2

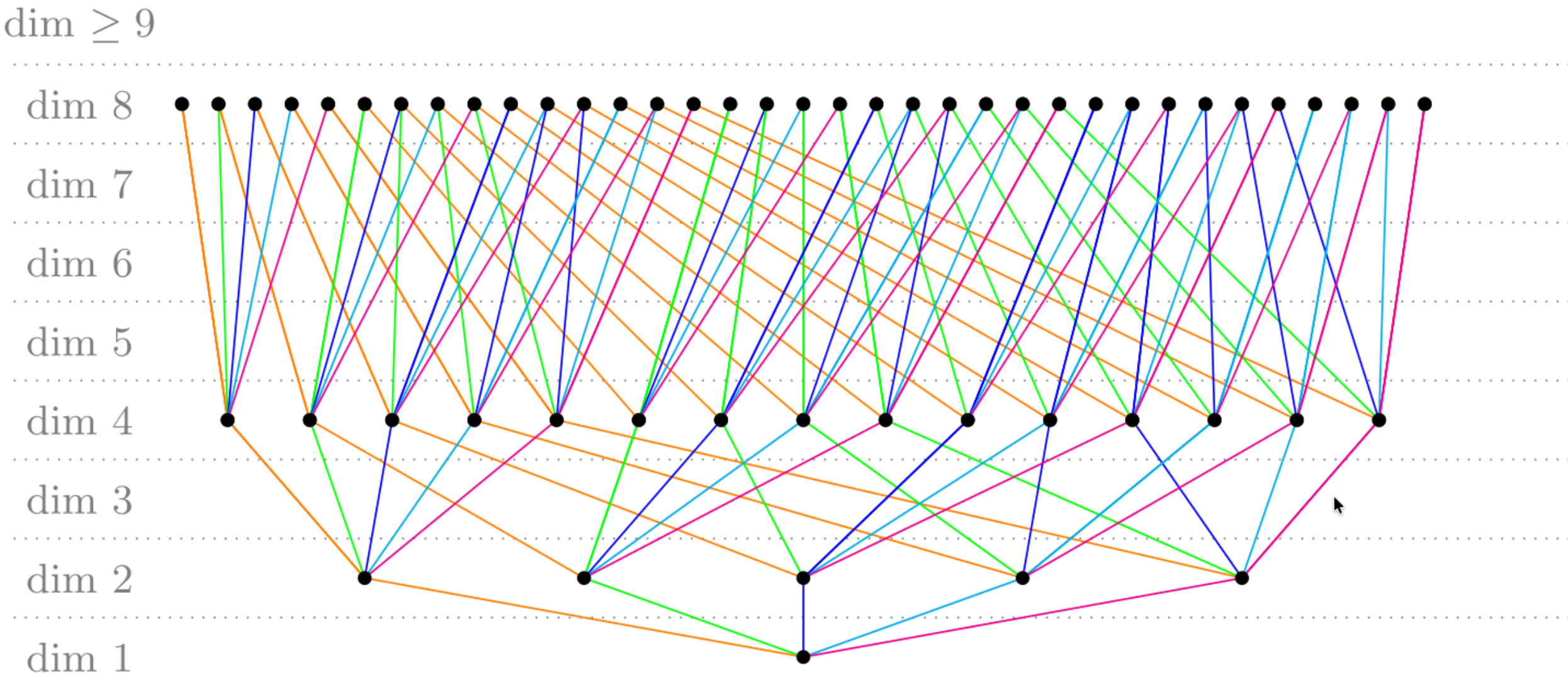


dimension 1



NB! The action of M_1 is not necessarily free on X_r for $r \geq 2$

KU LEUVEN



Cryptographic graded monoidal action,
quotient out by M_1 (or equivalently, for which $M_1 = \{1\}$)

KU LEUVEN

Some highlights of the cryptographic graded monoidal action

- **We formalise the need for certain algorithms (analogous to cryptographic group actions)**

KU LEUVEN

Some highlights of the cryptographic graded monoidal action

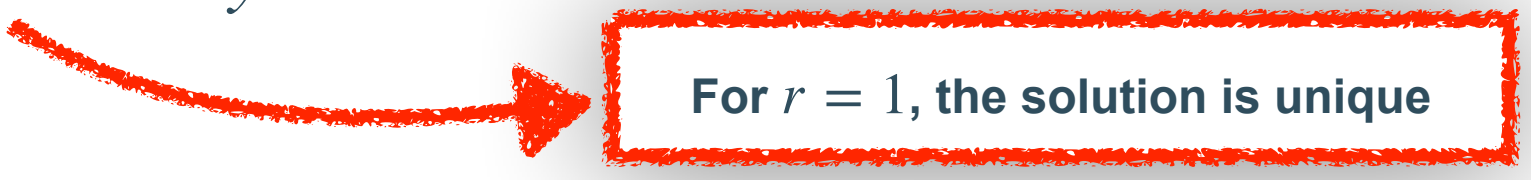
- **We formalise the need for certain algorithms (analogous to cryptographic group actions)**
- **Natural problems (DLOG, CDH, DDH) turn into families of problems:**

Some highlights of the cryptographic graded monoidal action

- **We formalise the need for certain algorithms (analogous to cryptographic group actions)**
- **Natural problems (DLOG, CDH, DDH) turn into families of problems:**
  - **e.g. (r, s)-graded vectorisation:**
    Given $x \in X_r$, $y := a \star x \in M_{rs}$, for some $a \in M_s$, find an $a' \in M_s$ such that $a' \star x = y$

Some highlights of the cryptographic graded monoidal action

- **We formalise the need for certain algorithms (analogous to cryptographic group actions)**
- **Natural problems (DLOG, CDH, DDH) turn into families of problems:**
  - **e.g. (r, s)-graded vectorisation:**
    Given $x \in X_r$, $y := a \star x \in M_{rs}$, for some $a \in M_s$, find an $a' \in M_s$ such that $a' \star x = y$



Some highlights of the cryptographic graded monoidal action

- **We formalise the need for certain algorithms (analogous to cryptographic group actions)**
- **Natural problems (DLOG, CDH, DDH) turn into families of problems:**
  - **e.g. (r, s)-graded vectorisation:**
    Given $x \in X_r$, $y := a \star x \in M_{rs}$, for some $a \in M_s$, find an $a' \in M_s$ such that $a' \star x = y$
- **Group action problems reduces to monoidal action problems.**
  - **e.g. vectorisation reduces to (r, s)-graded vectorisation for all r, s.**
  - **Hopefully strict gap.**

Some highlights of the cryptographic graded monoidal action

- **We formalise the need for certain algorithms (analogous to cryptographic group actions)**
- **Natural problems (DLOG, CDH, DDH) turn into families of problems:**
  - **e.g. (r, s)-graded vectorisation:**
    Given $x \in X_r$, $y := a \star x \in M_{rs}$, for some $a \in M_s$, find an $a' \in M_s$ such that $a' \star x = y$
- **Group action problems reduces to monoidal action problems.**
  - **e.g. vectorisation redu** **ion for all r, s.**
  - **Hopefully strict gap.**

Looking ahead:
[Robert] the general isogeny-problem reduces to $(1,2)$-graded vectorisation.

KU LEUVEN

New interesting problems

- (r, s)**-inversion: Given** $m \in M_s$**, and** $y := m \star x \in X_{rs}$**, find any** $x' \in X_s$ **satisfying** $y = m \star x'$

KU LEUVEN

New interesting problems

- **(r, s)-inversion: Given $m \in M_s$, and $y := m \star x \in X_{rs}$, find any $x' \in X_s$ satisfying $y = m \star x'$**
- **(Special case) (r, s)-extraction.**

KU LEUVEN

New interesting problems

- (r, s)**-inversion: Given** $m \in M_s$**, and** $y := m \star x \in X_{rs}$**, find any** $x' \in X_s$ **satisfying** $y = m \star x'$
- **(Special case)** (r, s)**-extraction.**
- (r, s)**-hiding: Let** $a \in M_r$ **be a secret element. Distinguish the sets**
  $S_0 := \mathrm{Im}(a \otimes_{r,s} -) = \{a \otimes b \mid b \in M_r\}$
  and
  $S_1 := \mathrm{Im}(- \otimes_{r,s} -) = \{a' \otimes b \mid a' \in M_s, b \in M_r\}$

KU LEUVEN

The Diffie-Hellman KEX v.4

Public parameters:

$\star : M \times X \to X$, **graded monoidal action (++)**

fixed element $x_0 \in X_1$

Alice:

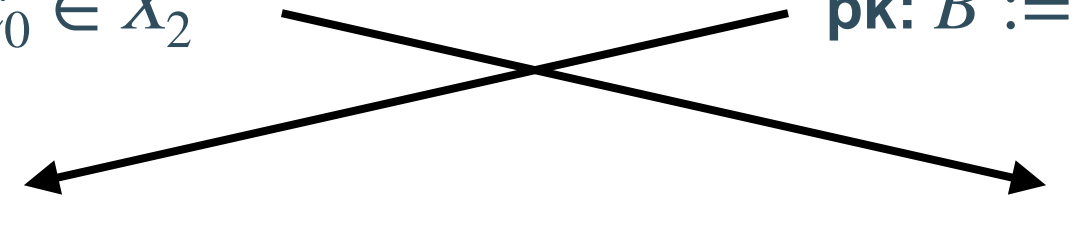
sk: $s_A \in M_2$

pk: $A := s_A \star x_0 \in X_2$

Bob:

sk: $s_B \in M_2$

pk: $B := s_B \star x_0 \in X_2$



B A

ss: $s_A \star B \in X_4$

ss: $s_B \star A \in X_4$

$$s_A \star B = (s_B \otimes s_A) \star x_0 = (s_A \otimes s_B) \star x_0 = s_B \star A$$

MIKE and the module action

Computing the module action in practice



To raise motivation, lets skip to the end...

Public parameters:

$E_0/\mathbb{F}_p : y^2 = x^3 - x$,

$p + 1 = 2^e \cdot f$, e **large,** $p \sim 256$ **bit**

$P, Q, \langle P, Q \rangle = E[2^e]$

Alice:

Bob:

*For practical reasons, we restrict to $s_A \mod 8 \in \{3,5\}$

To raise motivation, lets skip to the end...

Public parameters:

$E_0/\mathbb{F}_p : y^2 = x^3 - x$,

$p + 1 = 2^e \cdot f$, e **large,** $p \sim 256$ **bit**

$P, Q, \langle P, Q \rangle = E[2^e]$

Alice:

Bob:

sk*: $s_A \in \mathbb{Z}/2^e\mathbb{Z}$

pk: $E_A := E_0/\langle P + [s_A]Q \rangle$

*For practical reasons, we restrict to $s_A \mod 8 \in \{3,5\}$

To raise motivation, lets skip to the end...

Public parameters:

$E_0/\mathbb{F}_p : y^2 = x^3 - x$,

$p + 1 = 2^e \cdot f$, e **large,** $p \sim 256$ **bit**

$P, Q, \langle P, Q \rangle = E[2^e]$

Alice:

sk*: $s_A \in \mathbb{Z}/2^e\mathbb{Z}$

pk: $E_A := E_0/\langle P + [s_A]Q \rangle$

Bob:

sk*: $s_B \in \mathbb{Z}/2^e\mathbb{Z}$

pk: $E_B := E_0/\langle P + [s_B]Q \rangle$

*For practical reasons, we restrict to $s_A \mod 8 \in \{3,5\}$

To raise motivation, lets skip to the end...

Public parameters:

$E_0/\mathbb{F}_p : y^2 = x^3 - x$,

$p + 1 = 2^e \cdot f$, e **large,** $p \sim 256$ **bit**

$P, Q, \langle P, Q \rangle = E[2^e]$

Alice:

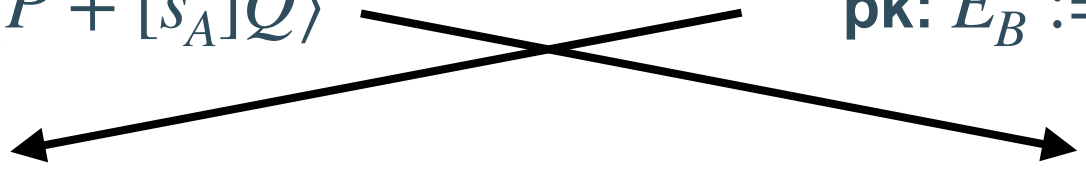
sk*: $s_A \in \mathbb{Z}/2^e\mathbb{Z}$

pk: $E_A := E_0/\langle P + [s_A]Q \rangle$

Bob:

sk*: $s_B \in \mathbb{Z}/2^e\mathbb{Z}$

pk: $E_B := E_0/\langle P + [s_B]Q \rangle$



E_B E_A

*For practical reasons, we restrict to $s_A \mod 8 \in \{3,5\}$

To raise motivation, lets skip to the end...

Public parameters:

$E_0/\mathbb{F}_p : y^2 = x^3 - x$,

$p + 1 = 2^e \cdot f$, e **large,** $p \sim 256$ **bit**

$P, Q, \langle P, Q \rangle = E[2^e]$

Alice:

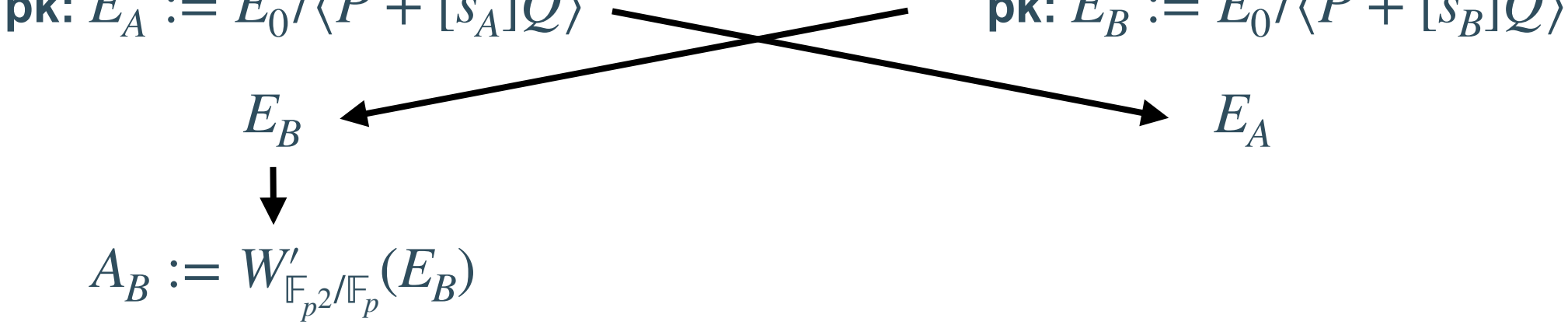
sk*: $s_A \in \mathbb{Z}/2^e\mathbb{Z}$

pk: $E_A := E_0/\langle P + [s_A]Q \rangle$

Bob:

sk*: $s_B \in \mathbb{Z}/2^e\mathbb{Z}$

pk: $E_B := E_0/\langle P + [s_B]Q \rangle$



*For practical reasons, we restrict to $s_A \mod 8 \in \{3,5\}$

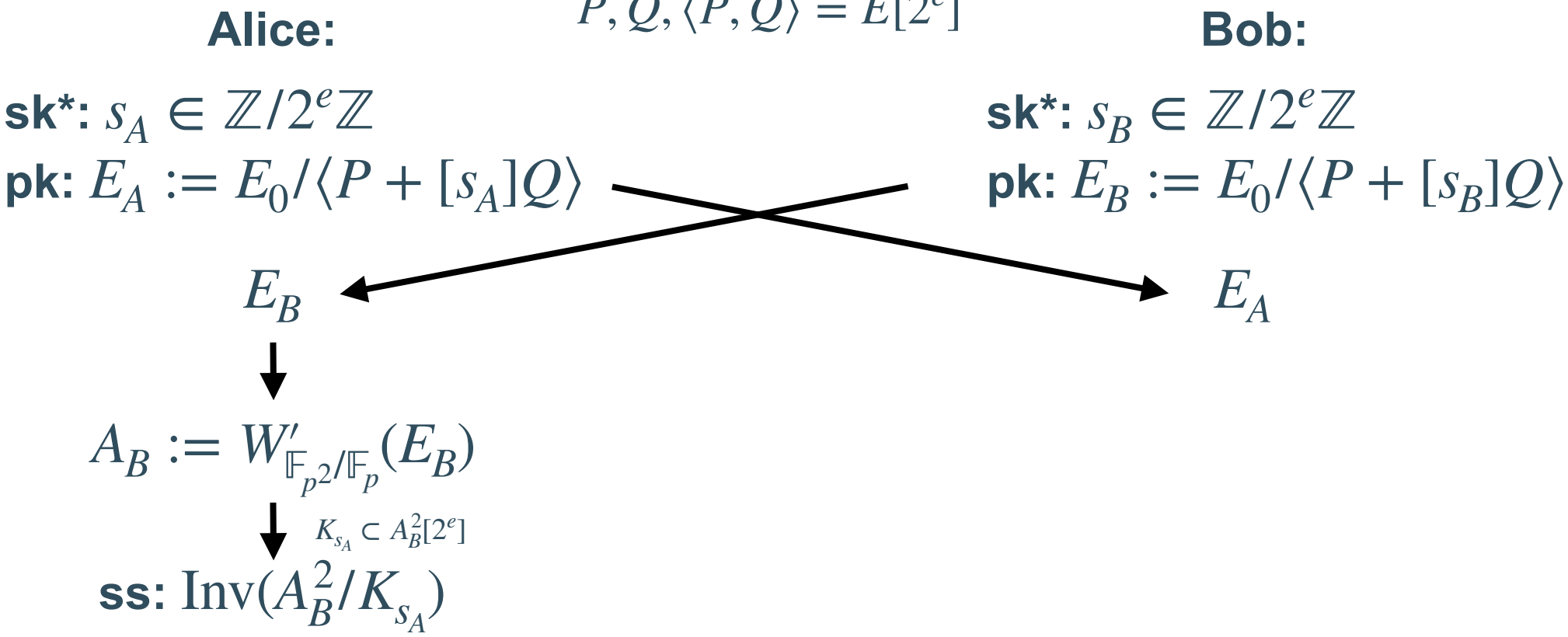
To raise motivation, lets skip to the end...

Public parameters:

$E_0/\mathbb{F}_p : y^2 = x^3 - x$,

$p + 1 = 2^e \cdot f$, e **large**, $p \sim 256$ **bit**

$P, Q, \langle P, Q \rangle = E[2^e]$



*For practical reasons, we restrict to $s_A \mod 8 \in \{3,5\}$

KU LEUVEN

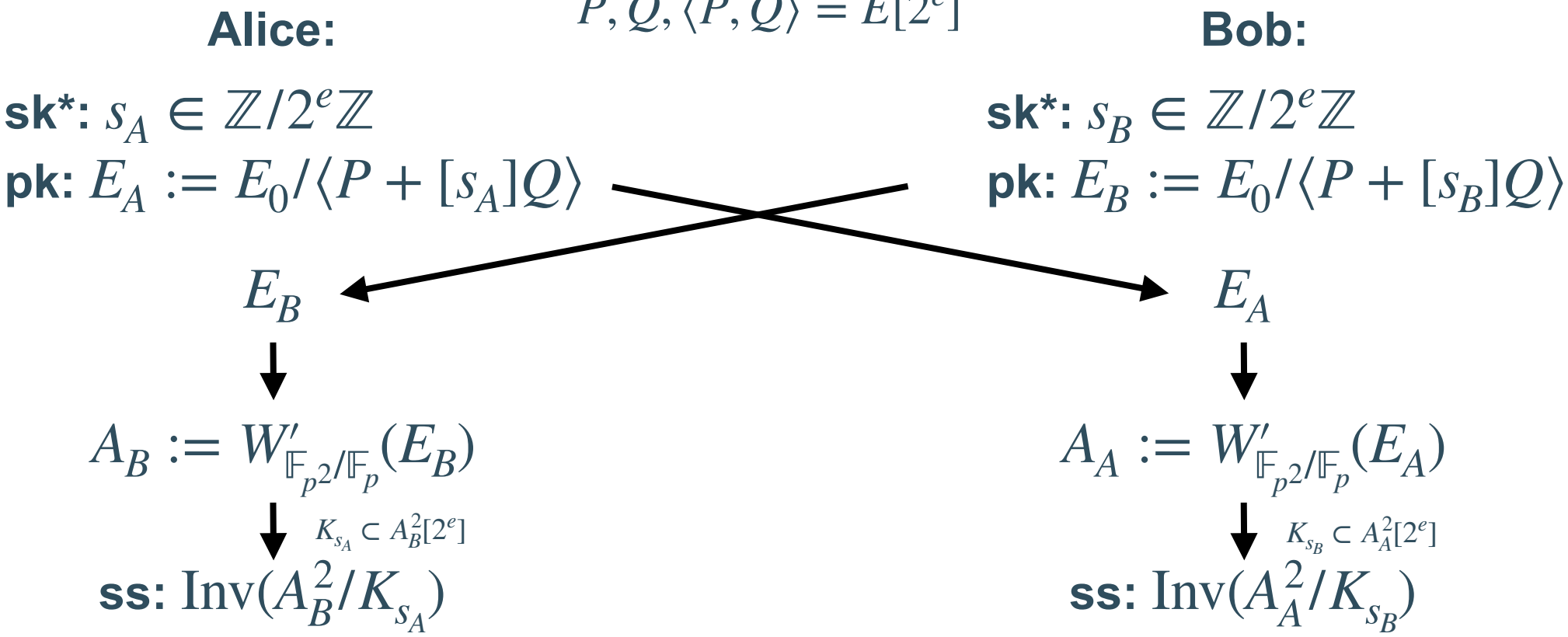
To raise motivation, lets skip to the end...

Public parameters:

$E_0/\mathbb{F}_p : y^2 = x^3 - x,$

$p + 1 = 2^e \cdot f$, e **large,** $p \sim 256$ **bit**

$P, Q, \langle P, Q \rangle = E[2^e]$



*For practical reasons, we restrict to $s_A \mod 8 \in \{3,5\}$

KU LEUVEN

Recall: the class group action

Context: R ring of integers* in an imaginary quadratic number field,
$E/\overline{\mathbb{F}_p}$ an elliptic curve with (primitive) embedding $\iota : R \hookrightarrow \mathrm{End}(E)$

*we will restrict to R maximal for simplicity, but this is not necessary, in particular, Damien's original paper does not restrict to this (In fact, it does not even restrict to quadratic over $\mathbb{Q}$).

KU LEUVEN

Recall: the class group action

Context: R ring of integers* in an imaginary quadratic number field, $E/\overline{\mathbb{F}_p}$ an elliptic curve with (primitive) embedding $\iota : R \hookrightarrow \mathrm{End}(E)$

Let $I(R)$ be the set of non-zero submodules $I \subset R$ (i.e. ideals).

- To any $I \in I(R)$, we can associate $E[I] := \{P \in E \mid r(P) = 0, \forall r \in I\}$, and the isogeny $\phi_I : E \to I \star_R E := E/E[I]$.

*we will restrict to R maximal for simplicity, but this is not necessary, in particular, Damien's original paper does not restrict to this (In fact, it does not even restrict to quadratic over $\mathbb{Q}$).

KU LEUVEN

Recall: the class group action

Context: R ring of integers* in an imaginary quadratic number field, $E/\overline{\mathbb{F}_p}$ an elliptic curve with (primitive) embedding $\iota : R \hookrightarrow \mathrm{End}(E)$

Let $I(R)$ be the set of non-zero submodules $I \subset R$ (i.e. ideals).

- To any $I \in I(R)$, we can associate $E[I] := \{P \in E \mid r(P) = 0, \forall r \in I\}$, and the isogeny $\phi_I : E \to I \star_R E := E/E[I]$.

Theorem: The association $I \to I \star_R E$ gives a well defined bijection between $Cl(R) \to Ell_p(R)$

While y'all know that the statement gets much stronger (we have a regular group action), lets forget that for now...

*we will restrict to R maximal for simplicity, but this is not necessary, in particular, Damien's original paper does not restrict to this (In fact, it does not even restrict to quadratic over $\mathbb{Q}$).

KU LEUVEN

Recall: the class group action

Context: R ring of integers* in an imaginary quadratic number field, $E/\overline{\mathbb{F}_p}$ an elliptic curve with (primitive) embedding $\iota : R \hookrightarrow \mathrm{End}(E)$

Let $I(R)$ be the set of non-zero submodules $I \subset R$ (i.e. ideals).

- To any $I \in I(R)$, we can associate $E[I] := \{P \in E \mid r(P) = 0, \forall r \in I\}$, and the isogeny $\phi_I : E \to I \star_R E := E/E[I]$.

Theorem: (Deuring) When $R = \mathcal{O}$ is a maximal quaternion order, The association $I \to I \star_{O_0} E$ gives a well defined bijection $Cls(\mathcal{O}) \to SS_p$ **d bijection between**

While y'all know that this statement gets much stronger (we have a regular group action), lets forget that for now...

*we will restrict to R maximal for simplicity, but this is not necessary, in particular, Damien's original paper does not restrict to this (In fact, it does not even restrict to quadratic over $\mathbb{Q}$).

KU LEUVEN

The module action

Context: R ring of integers* in an imaginary quadratic number field, $E/\overline{\mathbb{F}_p}$ an elliptic curve with (primitive) embedding $\iota : R \hookrightarrow \mathrm{End}(E)$

Let $I(R)M_n(R)$ be the set of non-zerofull-rank submodules $I \subset RM \subset R^n$

- To any $I \in I(R)M \subset M_n(R)$, we can associate $E[I] := \{P \in E \mid r(P) = 0, \forall r \in I\}$
  $E^n[M] := \{X \in E^n \mid m(X) = 0, \forall m \in M\}$, and the isogeny $\phi_I : E \to I \star_R E := E/E[I]$
  $\phi_M : E^n \to M \star_R A := E^n/E[M]$.

*we will restrict to R maximal for simplicity, but this is not necessary, in particular, Damien's original paper does not restrict to this (In fact, it does not even restrict to quadratic over $\mathbb{Q}$).

KU LEUVEN

The module action

Context: R ring of integers* in an imaginary qu

$E/\overline{\mathbb{F}_p}$ an elliptic curve with (primitive) em

Write $m \in M$ as $(r_1, \ldots, r_n) \in R^n$.

Then m defines a morphism

$$E^n \to E$$

$$(P_1, \ldots, P_n) \to \sum_i r_i(P_i)$$

Let $I(R)M_n(R)$ be the set of non-zero full-rank submodules $I \subset RM$

- To any $I \in I(R)M \subset M_n(R)$, we can associate $E[I] := \{P \in E \mid r(P) = 0, \forall r \in I\}$
  $E^n[M] := \{X \in E^n \mid m(X) = 0, \forall m \in M\}$, and the isogeny $\phi_I : E \to I \star_R E := E/E[I]$
  $\phi_M : E^n \to M \star_R A := E^n/E[M]$.

*we will restrict to R maximal for simplicity, but this is not necessary, in particular, Damien's original paper does not restrict to this (In fact, it does not even restrict to quadratic over $\mathbb{Q}$).

KU LEUVEN

The module action

Context: R ring of integers* in an imaginary quad
$E/\overline{\mathbb{F}_p}$ an elliptic curve with (primitive)

Let $I(R)M_n(R)$ be the set of non-zerofull-rank submodules $I \subset RM$

- To any $I \in I(R)M \subset M_n(R)$, we can associate $E[I] := \{P \in E \mid r(P) = 0, \forall r \in I\}$
  $E^n[M] := \{X \in E^n \mid m(X) = 0, \forall m \in M\}$, and the isogeny $\phi_I : E \to I \star_R E := E/E[I]$
  $\phi_M : E^n \to M \star_R A := E^n/$

Write $m \in M$ as $(r_1, \ldots, r_n) \in R^n$.

Then m defines a morphism

$$E^n \to E$$
$$(P_1, \ldots, P_n) \to \sum_i r_i(P_i)$$

Clearly, a generating set $\langle m_1, m_k \rangle = M$ is enough.
Writing $L_M = (r_{ij}) \in \mathbf{M}_{n,k}(R)$, with $m_i = r_{i1}, \ldots, r_{in}$ gives
$E^n[M] = \ker L_M$, where we view L_M as the morphism

$$L_M : E^n \to E^k$$
$$X \to L_M X$$

*we will restrict to R maximal for simplicity, but this is not necessary, in particular, Damien's original paper does not restrict to this (In fact, it does not even restrict to quadratic over $\mathbb{Q}$).

KU LEUVEN

The module action

Context: R ring of integers* in an imaginary quadratic number field,
$E/\overline{\mathbb{F}_p}$ an elliptic curve with (primitive) embedding $\iota : R \hookrightarrow \mathrm{End}(E)$

Let $I(R)M_n(R)$ be the set of non-zerofull-rank submodules $I \subset RM \subset R^n$

- To any $I \in I(R)M \subset M_n(R)$, we can associate $E[I] := \{P \in E \mid r(P) = 0, \forall r \in I\}$
  $E^n[M] := \{X \in E^n \mid m(X) = 0, \forall m \in M\}$, and the isogeny $\phi_I : E \to I \star_R E := E/E[I]$
  $\phi_M : E^n \to M \star_R A := E^n/E[M]$.

Theorem: ???

We discuss properties after looking at an example!

*we will restrict to R maximal for simplicity, but this is not necessary, in particular, Damien's original paper does not restrict to this (In fact, it does not even restrict to quadratic over $\mathbb{Q}$).

KU LEUVEN

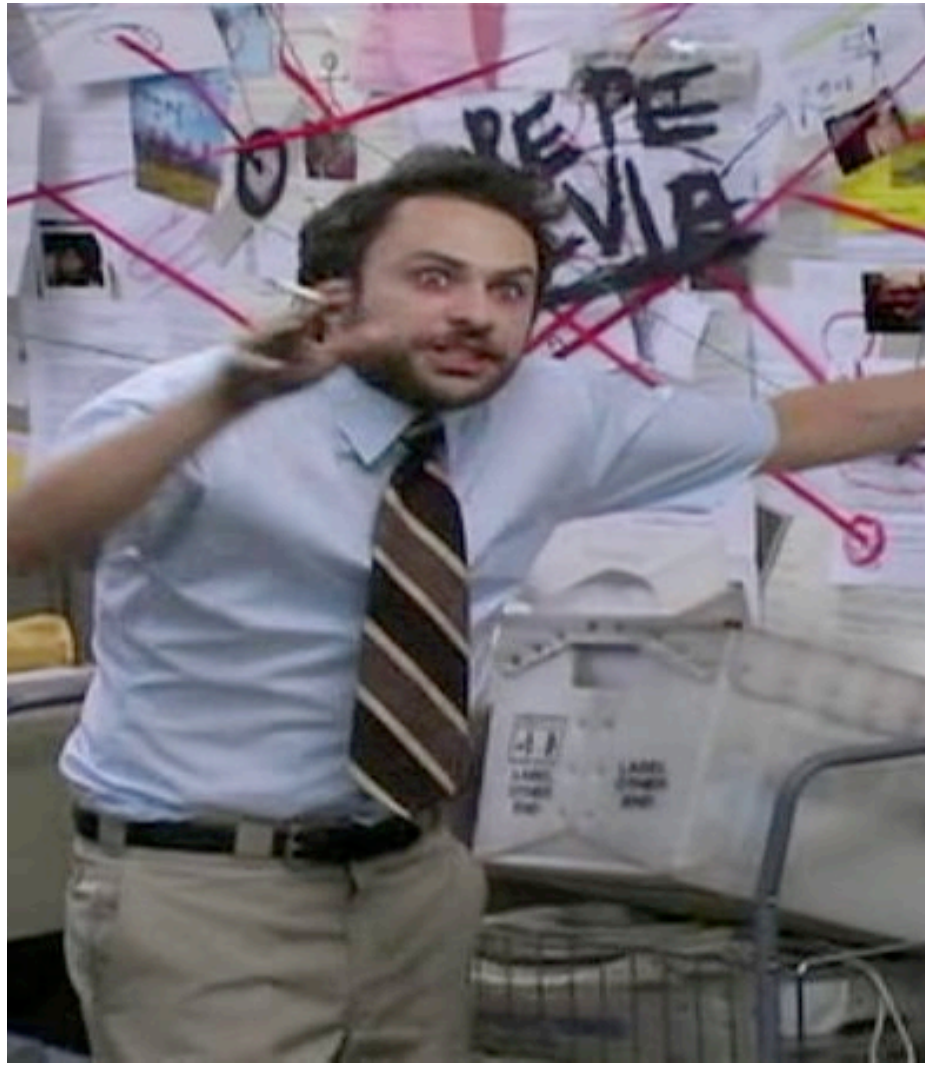
Example 3

Warning: Although quaternion orders comes a bit out of left-field here, I promise that studying this example is worth it.

KU LEUVEN

Some abstract non-sense

Properties of the module correspondence



This whole section will
happen on the board,
to make it possible to follow*

*The fact that I didn't have time to typeset it anyway, is a happy coincidence

The MIKE-protocol

Putting it all together + one final piece



We now have everything to make DH v.4!

Public parameters:

$E_0/\mathbb{F}_p : y^2 = x^3 - x$, **oriented by** $R = \mathbb{Z}[(1+\pi)/2]$

$p + 1 = 2^e \cdot f$, e **large,** $p \sim 256$ **bit**

$P, Q, \langle P, Q \rangle = E[2^e]$

Alice:

Bob:

We now have everything to make DH v.4!

Public parameters:

$E_0/\mathbb{F}_p : y^2 = x^3 - x$, **oriented by** $R = \mathbb{Z}[(1+\pi)/2]$

$p + 1 = 2^e \cdot f$, e **large,** $p \sim 256$ **bit**

$P, Q, \langle P, Q \rangle = E[2^e]$

Alice:

sk: $M_A \in \{M \subset R^2 \mid (M, H_R^n/\mathcal{O}) \textbf{ unimod.}\}$

pk: $A_A := M_A \star_R E_0$

Bob:

KU LEUVEN

We now have everything to make DH v.4!

Public parameters:

$E_0/\mathbb{F}_p : y^2 = x^3 - x$, **oriented by** $R = \mathbb{Z}[(1+\pi)/2]$

$p + 1 = 2^e \cdot f$, e **large,** $p \sim 256$ **bit**

$P, Q, \langle P, Q \rangle = E[2^e]$

Alice:

sk: $M_A \in \{M \subset R^2 \mid (M, H_R^n/\ell) \text{ unimod.}\}$

pk: $A_A := M_A \star_R E_0$

Bob:

sk: $M_B \in \{M \subset R^2 \mid (M, H_R^n/\ell) \text{ un}$

pk: $A_B := M_B \star_R E_0$

KU LEUVEN

We now have everything to make DH v.4!

Public parameters:

$E_0/\mathbb{F}_p : y^2 = x^3 - x$, **oriented by** $R = \mathbb{Z}[(1+\pi)/2]$

$p + 1 = 2^e \cdot f$, e **large,** $p \sim 256$ **bit**

$P, Q, \langle P, Q \rangle = E[2^e]$

Alice:

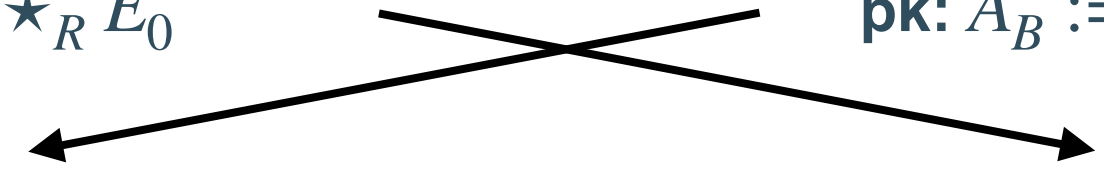
sk: $M_A \in \{M \subset R^2 \mid (M, H_R^n/\ell)$ **unimod.**$\}$

pk: $A_A := M_A \star_R E_0$

Bob:

sk: $M_B \in \{M \subset R^2 \mid (M, H_R^n/\ell)$ **un**

pk: $A_B := M_B \star_R E_0$



KU LEUVEN

We now have everything to make DH v.4!

Public parameters:

$E_0/\mathbb{F}_p : y^2 = x^3 - x$, **oriented by** $R = \mathbb{Z}[(1+\pi)/2]$

$p + 1 = 2^e \cdot f$, e **large,** $p \sim 256$ **bit**

$P, Q, \langle P, Q \rangle = E[2^e]$

Alice:

sk: $M_A \in \{M \subset R^2 \mid (M, H_R^n/\ell)$ **unimod.**$\}$

pk: $A_A := M_A \star_R E_0$

A_B

ss: $\mathrm{Inv}(M_A \star_R A_B)$

Bob:

sk: $M_B \in \{M \subset R^2 \mid (M, H_R^n/\ell)$ **un**

pk: $A_B := M_B \star_R E_0$

A_A

ss: $\mathrm{Inv}(M_B \star_R A_A)$

We now have everything to make DH v.4!

Public parameters:

$E_0/\mathbb{F}_p : y^2 = x^3 - x$, **oriented by** $R = \mathbb{Z}[(1+\pi)/2]$

$p + 1 = 2^e \cdot f$, e **large,** $p \sim 256$ **bit**

$P, Q, \langle P, Q \rangle = E[2^e]$

Alice:

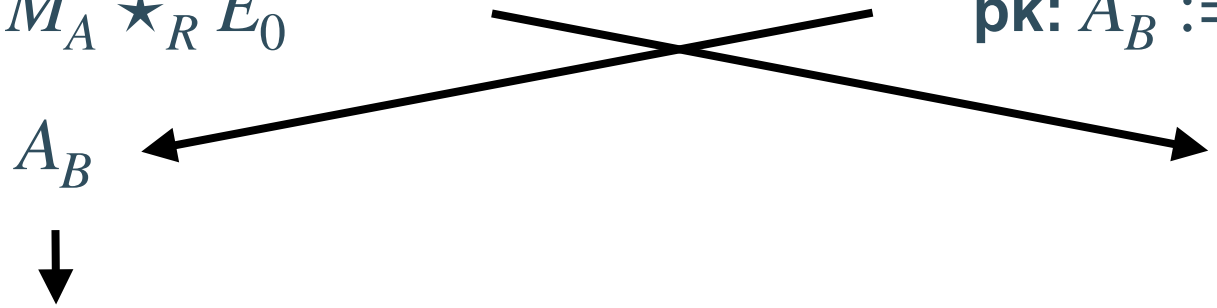
sk: $M_A \in \{M \subset R^2 \mid (M, H_R^n/\ell) \text{ unimod.}\}$

pk: $A_A := M_A \star_R E_0$

Bob:

sk: $M_B \in \{M \subset R^2 \mid (M, H_R^n/\ell)$ **un**

pk: $A_B := M_B \star_R E_0$



ss: $\mathrm{Inv}(M_A \star_R A_B)$

ss: $\mathrm{Inv}(M_B \star_R A_A)$

$$M_A \star A_B \cong (M_A \otimes M_B) \star_R E_0 \cong (M_B \otimes M_A) \star_R E_0 \cong M_B \star A_A$$

KU LEUVEN

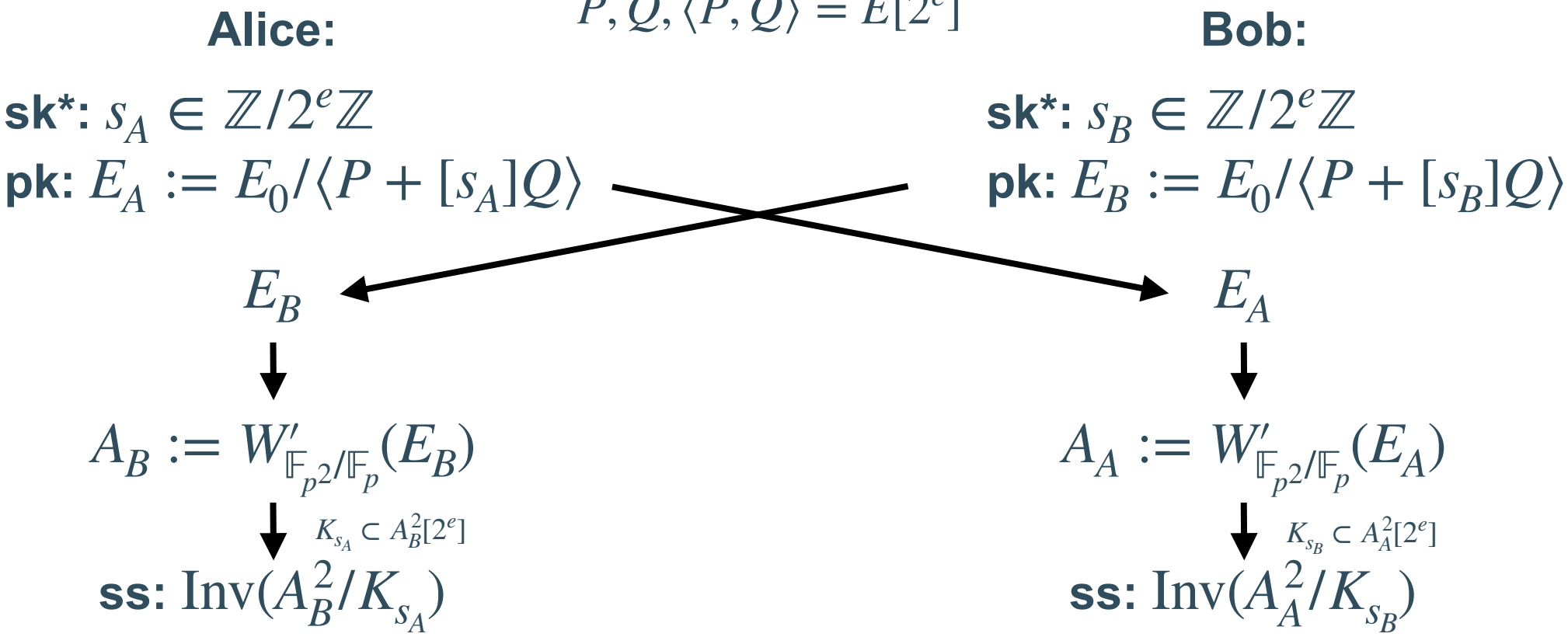
Hang on, this isn't what you showed before...

Public parameters:

$E_0/\mathbb{F}_p : y^2 = x^3 - x$, **oriented by** $R = \mathbb{Z}[(1+\pi)/2]$

$p + 1 = 2^e \cdot f$, e **large,** $p \sim 256$ **bit**

$P, Q, \langle P, Q \rangle = E[2^e]$



*For practical reasons, we restrict to $s_A \mod 8 \in \{3,5\}$

KU LEUVEN

One genius trick to take MIKE from good to great

Def: For $E/\mathbb{F}_{p^2}$, define $W'_{\mathbb{F}_{p^2}/\mathbb{F}_p}(E) := (E \times \pi(E))/\{(P, \pi(P)) \mid P \in E[2]\}$



Something very explicit and computable

KU LEUVEN

One genius trick to take MIKE from good to great

Def: For $E/\mathbb{F}_{p^2}$, define $W'_{\mathbb{F}_{p^2}/\mathbb{F}_p}(E) := (E \times \pi(E))/\{(P, \pi(P)) \mid P \in E[2]\}$

$E_0/\mathbb{F}_p$ supersingular, $\mathrm{End}(E_0) = \mathcal{O}_0$, oriented by $R = \mathbb{Z}[(1+\pi)/2]$

$$\iota : R \hookrightarrow \mathcal{O}_0$$

Theorem: For any ideal $I \subset \mathcal{O}_0$, we have

$$W'_{\mathbb{F}_{p^2}/\mathbb{F}_p}(I \star_{\mathcal{O}_0} E_0) \cong \mathrm{Res}_\iota(I) \star_R E_0$$

KU LEUVEN

One genius trick to take MIKE from good to great

Def: For $E/\mathbb{F}_{p^2}$, define $W'_{\mathbb{F}_{p^2}/\mathbb{F}_p}(E) := (E \times \pi(E))/\{(P, \pi(P)) \mid P \in E[2]\}$

$E_0/\mathbb{F}_p$ supersingular, $\mathrm{End}(E_0) = \mathcal{O}_0$, oriented by $R = \mathbb{Z}[(1+\pi)/2]$

$$\iota : R \hookrightarrow \mathcal{O}_0$$

Theorem: For any ideal $I \subset \mathcal{O}_0$, we have

$$W'_{\mathbb{F}_{p^2}/\mathbb{F}_p}(I \star_{\mathcal{O}_0} E_0) \cong \mathrm{Res}_\iota(I) \star_R E_0$$

Scholten's construction

KU LEUVEN

One genius trick to take MIKE from good to great

Def: For $E/\mathbb{F}_{p^2}$, define $W'_{\mathbb{F}_{p^2}/\mathbb{F}_p}(E) := (E \times \pi(E))/\{(P, \pi(P)) \mid P \in E[2]\}$

$E_0/\mathbb{F}_p$ supersingular, $\mathrm{End}(E_0) = \mathcal{O}_0$, oriented by $R = \mathbb{Z}[(1+\pi)/2]$

$$\iota : R \hookrightarrow \mathcal{O}_0$$

Theorem: For any ideal $I \subset \mathcal{O}_0$, we have

$$W'_{\mathbb{F}_{p^2}/\mathbb{F}_p}(I \star_{\mathcal{O}_0} E_0) \cong \mathrm{Res}_\iota(I) \star_R E_0$$

Scholten's construction

KU LEUVEN

One genius trick to take MIKE from good to great

Def: For $E/\mathbb{F}_{p^2}$, define $W'_{\mathbb{F}_{p^2}/\mathbb{F}_p}(E) := (E \times \pi(E))/\{(P, \pi(P)) \mid P \in E[2]\}$

$E_0/\mathbb{F}_p$ supersingular, $\mathrm{End}(E_0) = \mathcal{O}_0$, oriented by $R = \mathbb{Z}[(1+\pi)/2]$

$$\iota : R \hookrightarrow \mathcal{O}_0$$

Theorem: For any ideal $I \subset \mathcal{O}_0$, we have

$$W'_{\mathbb{F}_{p^2}/\mathbb{F}_p}(I \star_{\mathcal{O}_0} E_0) \cong \mathrm{Res}_\iota(I) \star_R E_0$$

Scholten's construction

Usual Deuring-correspondence computation

KU LEUVEN

One genius trick to take MIKE from good to great

Def: For $E/\mathbb{F}_{p^2}$, define $W'_{\mathbb{F}_{p^2}/\mathbb{F}_p}(E) := (E \times \pi(E))/\{(P, \pi(P)) \mid P \in E[2]\}$

$E_0/\mathbb{F}_p$ supersingular, $\mathrm{End}(E_0) = \mathcal{O}_0$, oriented by $R = \mathbb{Z}[(1+\pi)/2]$

$\iota : R \hookrightarrow \mathcal{O}_0$

Theorem: For any ideal $I \subset \mathcal{O}_0$, we have

$$W'_{\mathbb{F}_{p^2}/\mathbb{F}_p}(I \star_{\mathcal{O}_0} E_0) \cong \mathrm{Res}_\iota(I) \star_R E_0$$

Example 3

Scholten's construction

Usual Deuring-correspondence computation

KU LEUVEN

One genius trick to take MIKE from good to great

Theorem: For any ideal $I \subset \mathcal{O}_0$, we have

$$W'_{\mathbb{F}_{p^2}/\mathbb{F}_p}(I \star_{\mathcal{O}_0} E_0) \cong \mathrm{Res}_\iota(I) \star_R E_0$$

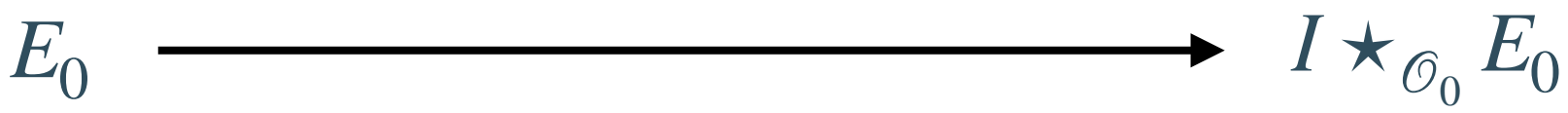
E_0

KU LEUVEN

One genius trick to take MIKE from good to great

Theorem: For any ideal $I \subset \mathcal{O}_0$, we have

$$W'_{\mathbb{F}_{p^2}/\mathbb{F}_p}(I \star_{\mathcal{O}_0} E_0) \cong \mathrm{Res}_\iota(I) \star_R E_0$$



KU LEUVEN

One genius trick to take MIKE from good to great

Theorem: For any ideal $I \subset \mathcal{O}_0$, we have

$$W'_{\mathbb{F}_{p^2}/\mathbb{F}_p}(I \star_{\mathcal{O}_0} E_0) \cong \mathrm{Res}_\iota(I) \star_R E_0$$



KU LEUVEN

One genius trick to take MIKE from good to great

Theorem: For any ideal $I \subset \mathcal{O}_0$, we have

$$W'_{\mathbb{F}_{p^2}/\mathbb{F}_p}(I \star_{\mathcal{O}_0} E_0) \cong \mathrm{Res}_\iota(I) \star_R E_0$$

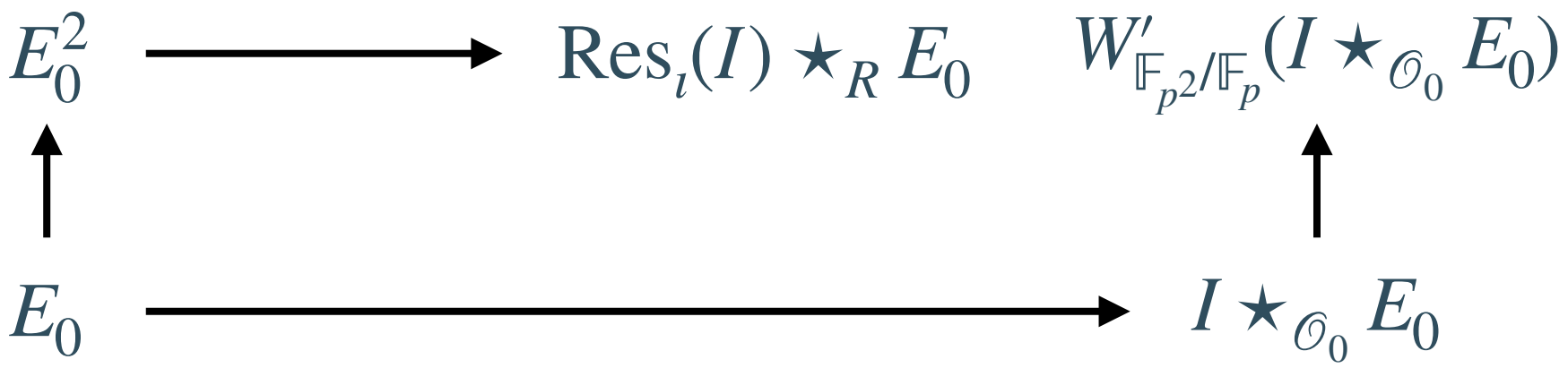


KU LEUVEN

One genius trick to take MIKE from good to great

Theorem: For any ideal $I \subset \mathcal{O}_0$, we have

$$W'_{\mathbb{F}_{p^2}/\mathbb{F}_p}(I \star_{\mathcal{O}_0} E_0) \cong \mathrm{Res}_\iota(I) \star_R E_0$$

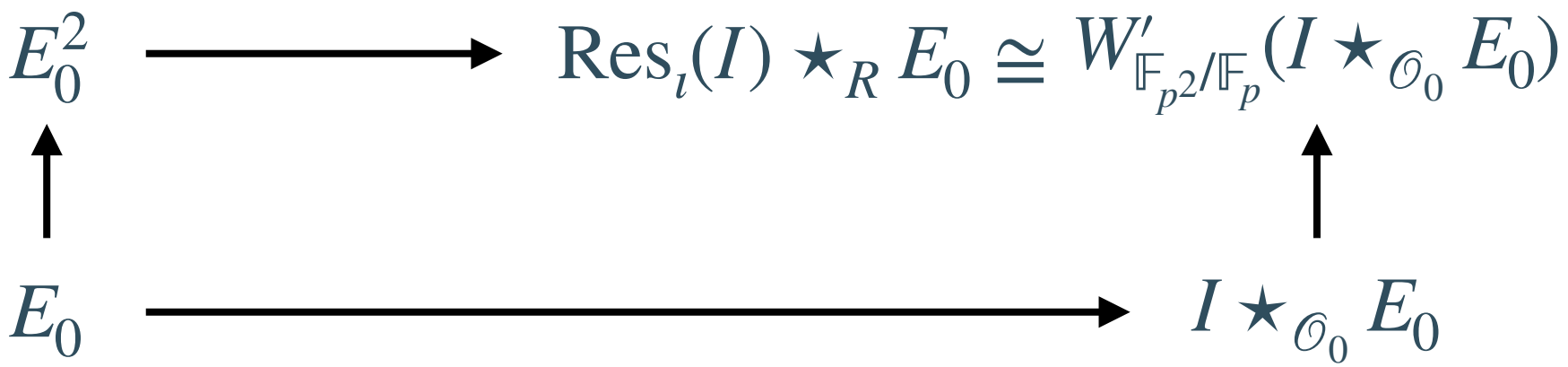


KU LEUVEN

One genius trick to take MIKE from good to great

Theorem: For any ideal $I \subset \mathcal{O}_0$, we have

$$W'_{\mathbb{F}_{p^2}/\mathbb{F}_p}(I \star_{\mathcal{O}_0} E_0) \cong \mathrm{Res}_\iota(I) \star_R E_0$$

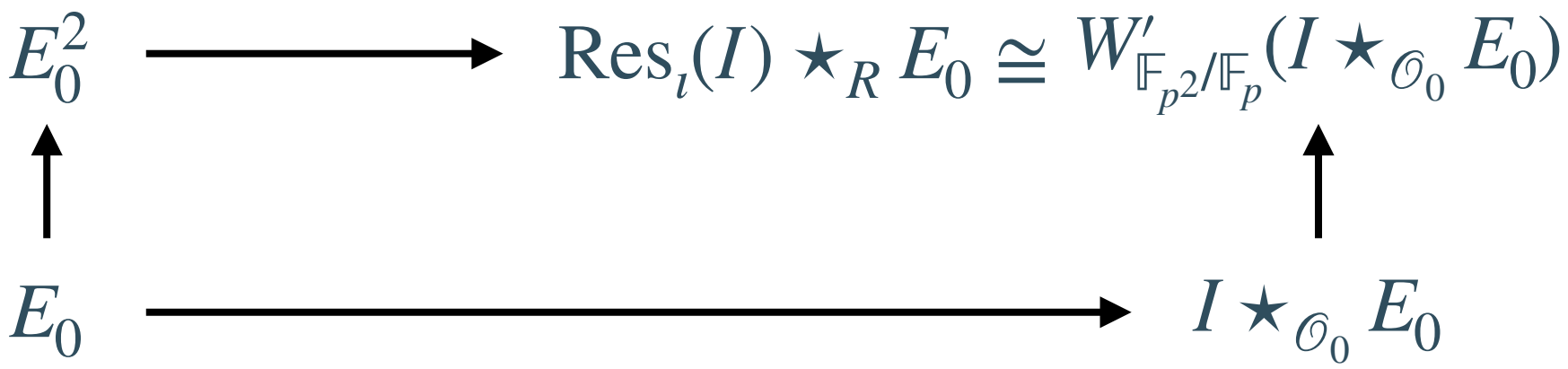


KU LEUVEN

One genius trick to take MIKE from good to great

Theorem: For any ideal $I \subset \mathcal{O}_0$, we have

$$W'_{\mathbb{F}_{p^2}/\mathbb{F}_p}(I \star_{\mathcal{O}_0} E_0) \cong \mathrm{Res}_\iota(I) \star_R E_0$$



Proof: Completely explicit by comparing kernels **(paper)**
Outline of abstract non-sense proof **(board)**

KU LEUVEN

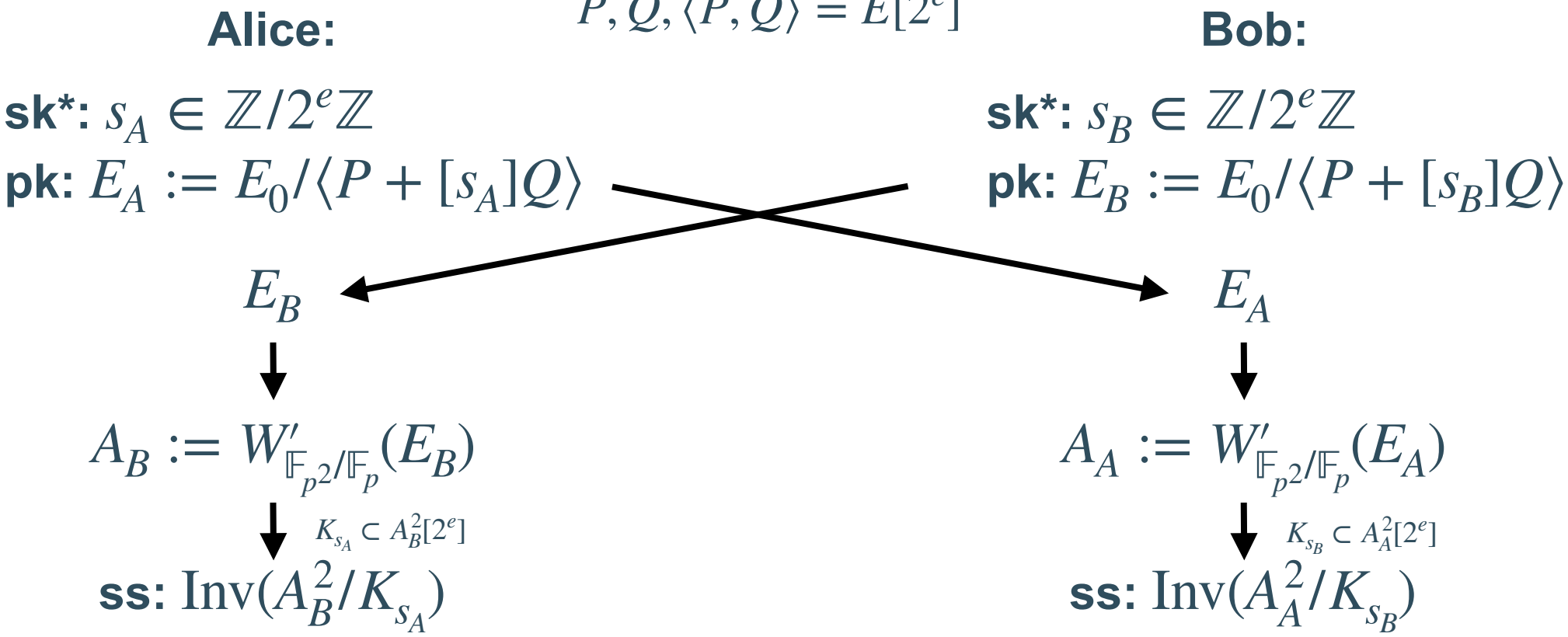
Finally, back where we started

Public parameters:

$E_0/\mathbb{F}_p : y^2 = x^3 - x$, **oriented by** $R = \mathbb{Z}[(1+\pi)/2]$

$p + 1 = 2^e \cdot f$, e **large,** $p \sim 256$ **bit**

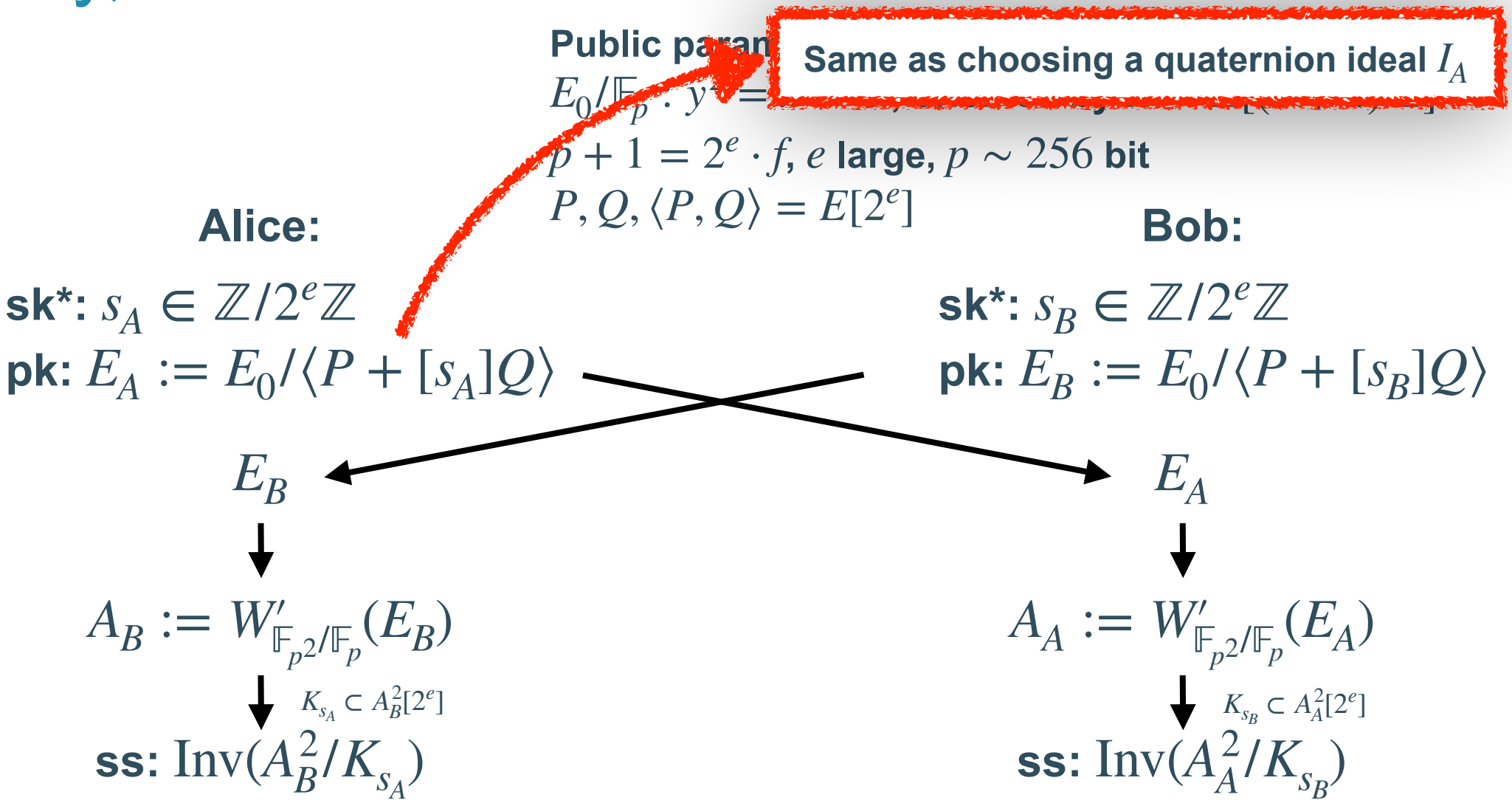
$P, Q, \langle P, Q \rangle = E[2^e]$



*For practical reasons, we restrict to $s_A \mod 8 \in \{3,5\}$

KU LEUVEN

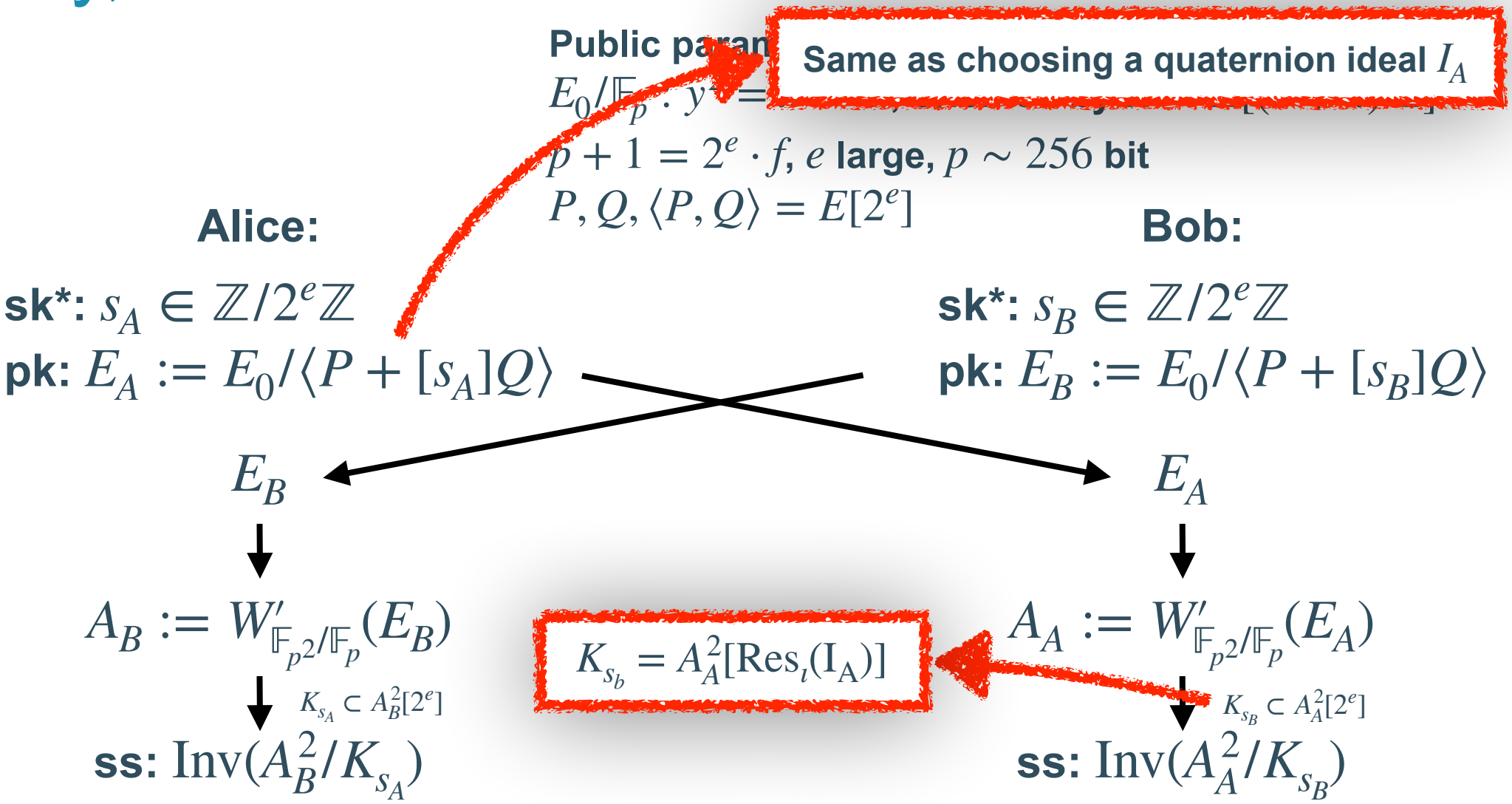
Finally, back where we started



*For practical reasons, we restrict to $s_A \mod 8 \in \{3,5\}$

KU LEUVEN

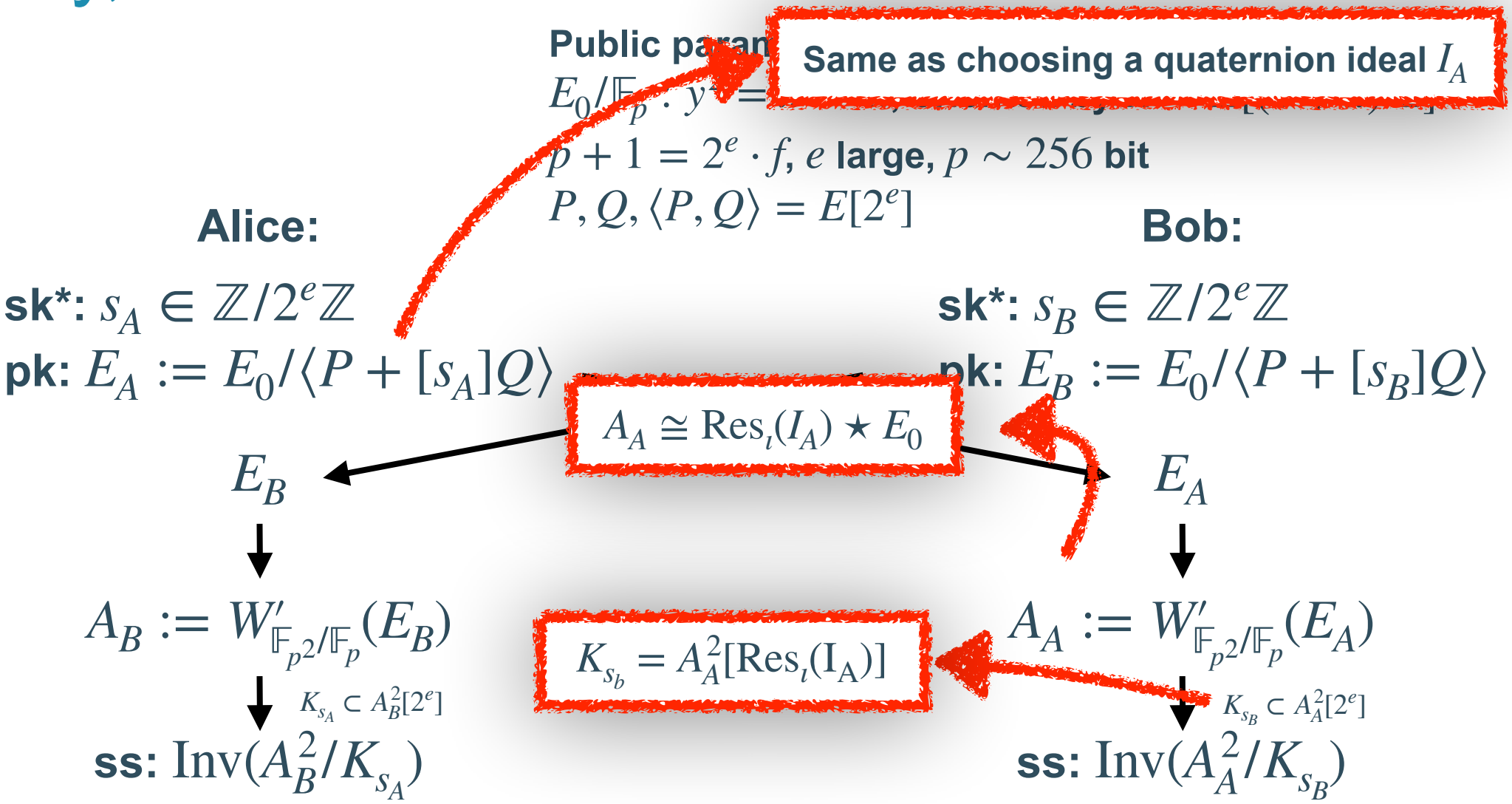
Finally, back where we started



*For practical reasons, we restrict to $s_A \mod 8 \in \{3,5\}$

KU LEUVEN

Finally, back where we started



*For practical reasons, we restrict to $s_A \mod 8 \in \{3,5\}$

KU LEUVEN

Further topics

These topics deserve a presentation of their own (by someone else than me...)



Abelian surfaces of Weil-type

- All the 4-dimensional ppav's we encounter are extremely special
- From my limited understanding, highlights are:
  - They all have a $\mathbb{Z}[i]$-orientation
  - Multiplication by i leads to many symmetries we exploit for incredible formulas for isogenies!
  - While complete invariants for ppav's of dimension 4 is unsolved, we have managed to find complete invariants for these special varieties!!

KU LEUVEN

Example 3.

Didn't have time to write this down...

Other highlights to look forward to:

- **Gluing computations are fast and constant time (as is the rest of MIKE)**
- **Formalising security**

All this and more is soon coming to an eprint near you!

MIKE: a fast and compact post-quantum NIKE

Andrea Basso[1], Pierrick Dartois[2], Max Duparc[3], Jonathan Komada Eriksen[4], Sabrina Kunzweiler[5], Michael Meyer, Giacomo Pope[6], Krijn Reijnders[7], Damien Robert[6], Ryan Rueger, and Sina Schaeffler

Abstract non-sense

Ask if anyone wants a recap of essential prelims:

- Def. of a (locally small) category
- — " — functor
- — " — pair of adjoint functors
- — " — (anti) equivalence of cats.
- Yoneda's lemma

Observe that for class-group action, $I * E_0$ satisfies (→ $\mathfrak{o}$-oriented morphisms)

$$\mathrm{Hom}(X, E_I) = \mathrm{Hom}_R(I, \mathrm{Hom}(X, E_0))$$

for all $X \in \mathcal{E}\ell\ell_p(R)$.

(Assume $X = J * E_0$. Then $\mathrm{Hom}(X, E_0) = J$, and $\mathrm{Hom}_R(I, J) = (I : J) = I^{-1}J = \mathrm{Hom}(X, E_0)$)

Take this as (Yoneda) def! [if it exists]

$M * E_0$ is the object satisfying

$$\mathrm{Hom}(X, M * E_0) = \mathrm{Hom}_R(M, \mathrm{Hom}(X, E_0))$$

- This def. requires proof that $M * E$ actually exists  believe

Prop: Let A be R-oriented proper g.s., and let M_1, M_2 be f.g. R-mods. Then

$$M_1 * (M_2 * A) = (M_1 \otimes M_2) * A.$$

Proof: Hom-tensor adjunction.

$$\begin{aligned}
&\mathrm{Hom}(X, M_1 * (M_2 * A)) \\
&\overset{\text{def.}}{=} \mathrm{Hom}_R(M_1, \mathrm{Hom}(X, M_2 * A)) \\
&\overset{\text{def}}{=} \mathrm{Hom}_R(M_1, \mathrm{Hom}_R(M_2, \mathrm{Hom}(X, A))) \\
&\overset{\text{adj.}}{=} \mathrm{Hom}_R(M_1 \otimes M_2, \mathrm{Hom}(X, A)) \\
&\overset{\text{def.}}{=} \mathrm{Hom}(X, (M_1 \otimes M_2) * A)
\end{aligned}$$

□

Note: Once we show that this construction agrees with the other one for ideals $I \subset R$, this immidiately gives the class grp. action!

The big theorem

Thm:

The functor

F : f.p. R-mod $\to$ (R-oriented) proper g.s.

$M \to M * E_0$

NB! critical that E_0 is primitively R-oriented

is exact and faithful.

Restricted to torsion-free f.p. R-md it induces an antiequivalence of categories to $Ab_{E_0,R}$

↗

R-oriented ab. vars, R-isogenous to E_0^n for some $n \in \mathbb{N}$.

(This thm is of course far from easy, but we will now use it to derive other stuff.)

We have defined $M * E_0$ in two ways when $M \subseteq R^n$ (slides vs. board).

Outline why these agree:

- In the proof that $M * E_0$ (board) exists, one explicitly constructs it by taking f.p.

$$R^n \xrightarrow{L_M} R^n \to M \to 0$$

$$\updownarrow$$

$$0 \to M * A \to A^n \xrightarrow{L_M} A^n$$

- The earlier construction is explained by

$$0 \to M \to R^n \to R^n/M \to 0$$

$$\updownarrow$$

$$0 \to R^n/M * A \to A^n \to M * A \to 0$$

We instead recover this, the kernel of $\phi : A^n \to M * A$, and use it to compute $M * A$.

Tie back to graded monoidal actions:

Fact: Projective (= torsion free) R-modules (R Dedekind) form a monoid under the tensor product.
Rank turns it into a graded monoid.

Thus (grading given by dimension) $M = \{P \mid P \text{ proj. f.p. } R\text{-mod}\}/\cong^{ism}$ acts on $X = \{A \in Ab_{\aleph_0,R}\}/\sim$

Further:
- Rankwise finite: Classical
- Base free: Fully faithful (eq. of cats)
- Base transitive: Essentially surjective

However: (Steinitz)
Thm: $M = \mathbb{N} \times Cl(R)$ as sets

NB! Not as monoids. Mult is given by
$(n, [I]) \otimes (n, [J]) = (mn, [J]^m [I]^n)$

So it seems we've gained very little...

Duality, polarizations, and Hermitian forms

Instead of simply looking at R-oriented Ab. vars., we want to look at R-oriented pp av's!

(A, λ_A, i)

Requirement:

$i : R \hookrightarrow \mathrm{End}_{\mathrm{sf}}(A)$ is an injection of rings with involution, i.e. $i(\bar{r}) = i(r)^\dagger$.

We skip the def. of a Hermitian form. All we need is

Fact: $H_{\mathbb{R}^n} : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$

$$((r_1, \dots, r_n), (r_1', \dots, r_n')) \to \sum r_i \bar{r_i}'$$

An example of one on $\mathbb{R}^n$.

Thm: $M \subset \mathbb{R}^n$ full rank, $(M : \mathbb{Z}^n) = \ell^n$.

Let (A, λ_A, i) be as before. Then

$H_{\mathbb{R}^n/\ell} : M \times M \to \mathbb{R}$ is unimodular iff. $A^n[M]$ is maximally isotropic.

For this pres: Ignore this, think of this

Corollary: Same setup as before:

$(M, H_{\mathbb{R}^n/\ell})$ unimodular

$\Rightarrow$ $M * A$ is principally polarised

(or rather λ_A induces a prin. polarisation on $M * A$).

More generally, the above is true for (M, H_M) unimodular (not necessarily $M \subset \mathbb{R}^n$ sublattice)

Corollary: Now let $M' := \{(P, H) \mid P \in \text{f.g. proj. } R\text{-mod}, H : P \times P \to R \text{ unimod Hermitian}\}$ / isometry

And $X' := \{(A, \lambda_A, i) \mid (A, i) \in Ab_{E_0, R}, \lambda_A \text{ p.p.}, i : R \hookrightarrow \text{End}(A) \text{ Ring* hom}\}$ / $\cong$

Then - M' is a graded monoid (under tensor). pol. ison

- $M' \times X' \to X'$ is a graded monoidal action.
- still satisfies all nice prop

Weil restriction

(1) $W'_{\mathbb{F}_{p^i}/\mathbb{F}_p}(I *_{\mathcal{O}_0} E_0) = (\mathcal{O}_0 \otimes_R \mathrm{res}_i(I)) *_{\mathcal{O}_0} E_0$

↳ Skip this

(2) For any R-module M, $(\mathcal{O}_0 \otimes M) *_{\mathcal{O}_0} E_0 \cong (M *_R E_0)$ (forgetting about R-orientation)

Proof: $\mathrm{Hom}_{Ab}(X, M *_R E_0)$

$\overset{\text{def}}{=} \mathrm{Hom}_R(M, \mathrm{Hom}_{Ab}(X, E_0))$

$= \mathrm{Hom}_{\mathcal{O}_0}(\mathcal{O}_0 \otimes M, \mathrm{Hom}_{Ab}(X, E_0))$

(adj.)

$\overset{\text{def.}}{=} \mathrm{Hom}_{Ab}(X, (\mathcal{O}_0 \otimes M) *_{\mathcal{O}_0} E_0)$

↳ $(\mathcal{O}_0 \otimes -, \mathrm{Res}_i)$ are adjoint

(very similar to tensor/hom adjunction, though we swap categories).